

Reference Schematics For RK3588S

RK3588S_Tablet_Demo_SCH

Main Functions Introduction

- 1) Charger: 1Cell Battery_QC
- 2) PMIC: 1 x RK806-1+DiscretePower
- 3) RAM: 2 x 32bits LPDDR4/4x
- 4) ROM: eMMC5.1(Default)
- 5) Support: 1 x Type-C 3.0(with DP function)
- 6) Support: 1 x 4Lanes MIPI D/CPHY RX Camera
- 7) Support: 1 x 2Lanes MIPI DPHY RX Camera
- 8) Support: 1 x 4Lanes MIPI D/CPHY TX
- 9) Support: a/b/g/n/ac/ax 2T2R WIFI(PCIE) + BT5.0
- 11) Support: 1 x Headphone + 2 x Speaker out + 1 x Analog MIC
- 12) Support: 2 x PDM MIC Array
- 13) Support: Gyroscope+G-sensor+Ambient Light+Proximity

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Project:	RK3588S_Demo		
File:	00.Cover Page		
Date:	Friday, January 07, 2022		Rev: V10
Designed by:	Joseph	Reviewed by:	<Checker>
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Note

The power suffix S0 or S3 means:

S3: Keep power On during sleeping

S0:Power off during sleeping

Generate Bill of Materials

Header:

Item\tPart\tDescription\tPCB Footprint\tReference\tQuantity\tOption

Combined property string:

{Item}\t{Value}\t{Description}\t{PCB Footprint}\t{Reference}\t{Quantity}\t{Option}

Description

Note

Option

Notes

NOTE 1:
Component parameter description
1. DNP stands for component not mounted temporarily
2. If Value or option is DNP, which means the area is reserved without being mounted

NOTE 2:
Please use our recommended components to avoid too many changes.
For more informations about the second source,please refer to our AVL.

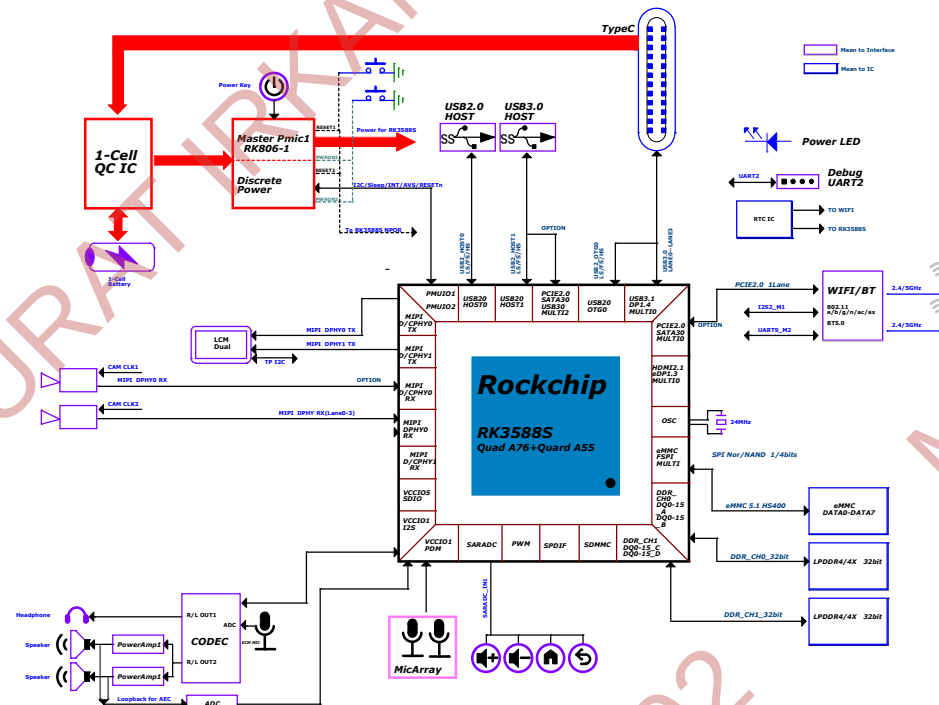
Revision History

Version	Date	By	Change Dscription	Approved
V1.0	2022-01-05	Joseph.Wei	1:Revision preliminary version	
V1.1	2022-02-18	Joseph.Wei	1.C1604,C1612的电容改成1uF/4V。 2.为了减少待机功耗，将PMUIO2电源域改成1.8V，此IO域对应外设IO电压相应修改 3.把L2203，L2205，L2207，L2300，L2301，L2302电感由0.22uH(TDK)改为0.24uH(Sunlord)；L2201的电感由0.22uH(TDK)改为0.22uH (Sunlord)，封装IND_404020。	

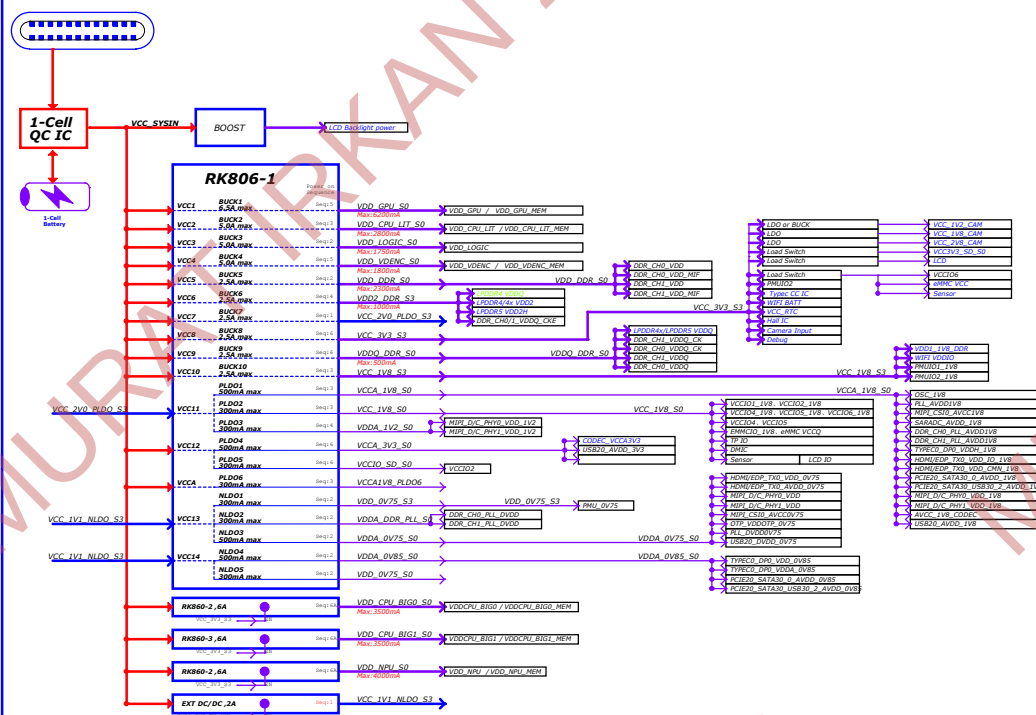
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Project:	RK3588S_Demo		
File:	02.Revision History		
Date:	Wednesday, February 23, 2022		Rev: V10
Designed by:	Joseph	Reviewed by:	<Checker> Sheet: 3 of 32

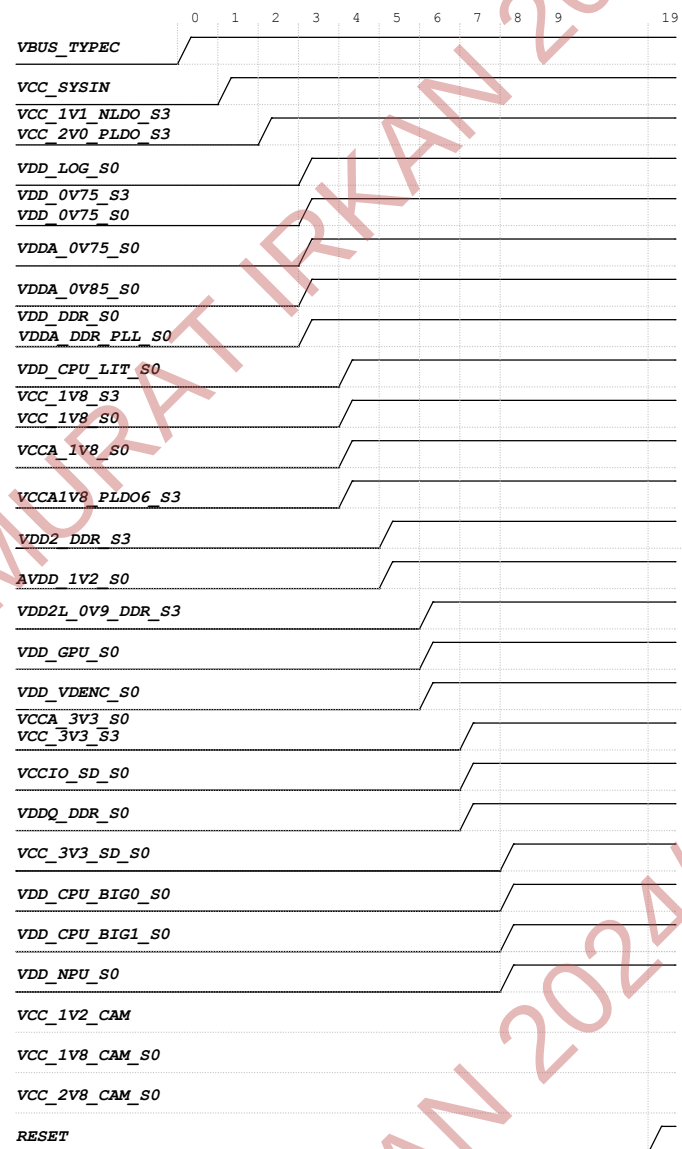
RK3588S Tablet Demo Block Diagram for 1-Cell Charger



Power tree for 1-Cell Charger



Power Sequence



Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Sleep ON/OFF	Peak Current	Sleep Current
VCC_SYSIN	RK806-1_BUCK1	6.5A	VDD_GPU_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK2	5A	VDD_CPU_LIT_S0	Slot:3	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK3	5A	VDD_LOG_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK4	3A	VDD_VDENC_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK5	2.5A	VDD_DDR_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK6	2.5A	VDD2_DDR_S3	Slot:4	ADJ FB=0.5V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK7	2.5A	VCC_2V0_PLDO_S3	Slot:1	2.0V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK8	2.5A	VCC_3V3_S3	Slot:6	3.3V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK9	2.5A	VDDQ_DDR_S0	Slot:6	ADJ FB=0.5V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK10	2.5A	VCC_1V8_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_2V0_PLDO_S3	RK806-1_PLDO1	0.5A	VCC_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO2	0.3A	VCCA_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO3	0.3A	VDDA_1V2_S0	Slot:4	1.2V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_PLDO4	0.5A	VCCA_3V3_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO5	0.3A	VCCIO_SD_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO6	0.3A	VCCA1V8_PLDO6_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_1V1_NLDO_S3	RK806-1_NLDO1	0.3A	VDD_0V75_S3	Slot:2	0.75V	ON	ON	TBD	TBD
	RK806-1_NLDO2	0.3A	VDDA_DDR_PLL_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO3	0.5A	VDDA_0V75_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_1V1_NLDO_S3	RK806-1_NLDO4	0.5A	VDDA_0V85_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO5	0.3A	VDD_0V75_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-2	6A	VDD_CPU_BIG0_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-3	6A	VDD_CPU_BIG1_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	EXT BUCK	2A	VCC_1V1_NLDO_S3	Slot:1	1.1V	ON	ON	TBD	TBD
VCC_SYSIN	EXT BUCK	2A	VDD2L_0V9_DDR_S3	Slot:5	0.9V	ON	ON	TBD	TBD
VCC_SYSIN	EXT BUCK	2.5A	VCC_3V3_SD_S0	Slot:6A	3.3V	ON	OFF	TBD	TBD
VCC_SYSIN	EXT_BUCK or LDO	2A	VCC_1V2_CAM_S0	OFF	1.2V	OFF	OFF	TBD	TBD
VCC_SYSIN	LDO	0.5A	VCC_1V8_CAM_S0	OFF	1.8V	OFF	OFF	TBD	TBD
VCC_SYSIN	LDO	0.5A	VCC_2V8_CAM_S0	OFF	2.8V	OFF	OFF	TBD	TBD

IO Power Domain Map

IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	Operating Voltage
PMUIO1	Pin N36 N37	1.8V Only	PMUIO1_1V8	VCC_1V8_S3	1.8V
PMUIO2	Pin V37 Y37	1.8V or 3.3V	PMUIO2_1V8	VCC_1V8_S3	1.8V
EMMCIO	Pin V35 V36	1.8V Only	PMUIO2	VCC_1V8_S3	1.8V
	Pin AC35	1.8V Only	EMMCIO_1V8	VCC_1V8_S0	1.8V
	Pin AC36	1.8V Only			
VCCIO1	Pin H31	1.8V Only	VCCIO1_1V8	VCC_1V8_S0	1.8V
VCCIO2	Pin AK11	1.8V or 3.3V	VCCIO2_1V8	VCC_1V8_S0	1.8V
	Pin AK10	1.8V or 3.3V	VCCIO2	VCC_1V8_S0	1.8V
VCCIO4	Pin G27 G28	1.8V or 3.3V	VCCIO4_1V8	VCC_1V8_S0	1.8V
	Pin G31	1.8V or 3.3V	VCCIO4	VCC_1V8_S0	1.8V
VCCIO5	Pin AF35 AF36	1.8V or 3.3V	VCCIO5_1V8	VCC_1V8_S0	1.8V
	Pin AC33 AC34	1.8V or 3.3V	VCCIO5	VCC_1V8_S0	1.8V
VCCIO6	Pin A134	1.8V or 3.3V	VCCIO6_1V8	VCC_1V8_S0	1.8V
	Pin A133 AM33	1.8V or 3.3V	VCCIO6	VCC_3V3_S0	3.3V

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Project: RK3588S_Demo

File: 05.System Power Sequence

Date: Monday, January 24, 2022

Designed by: Joseph

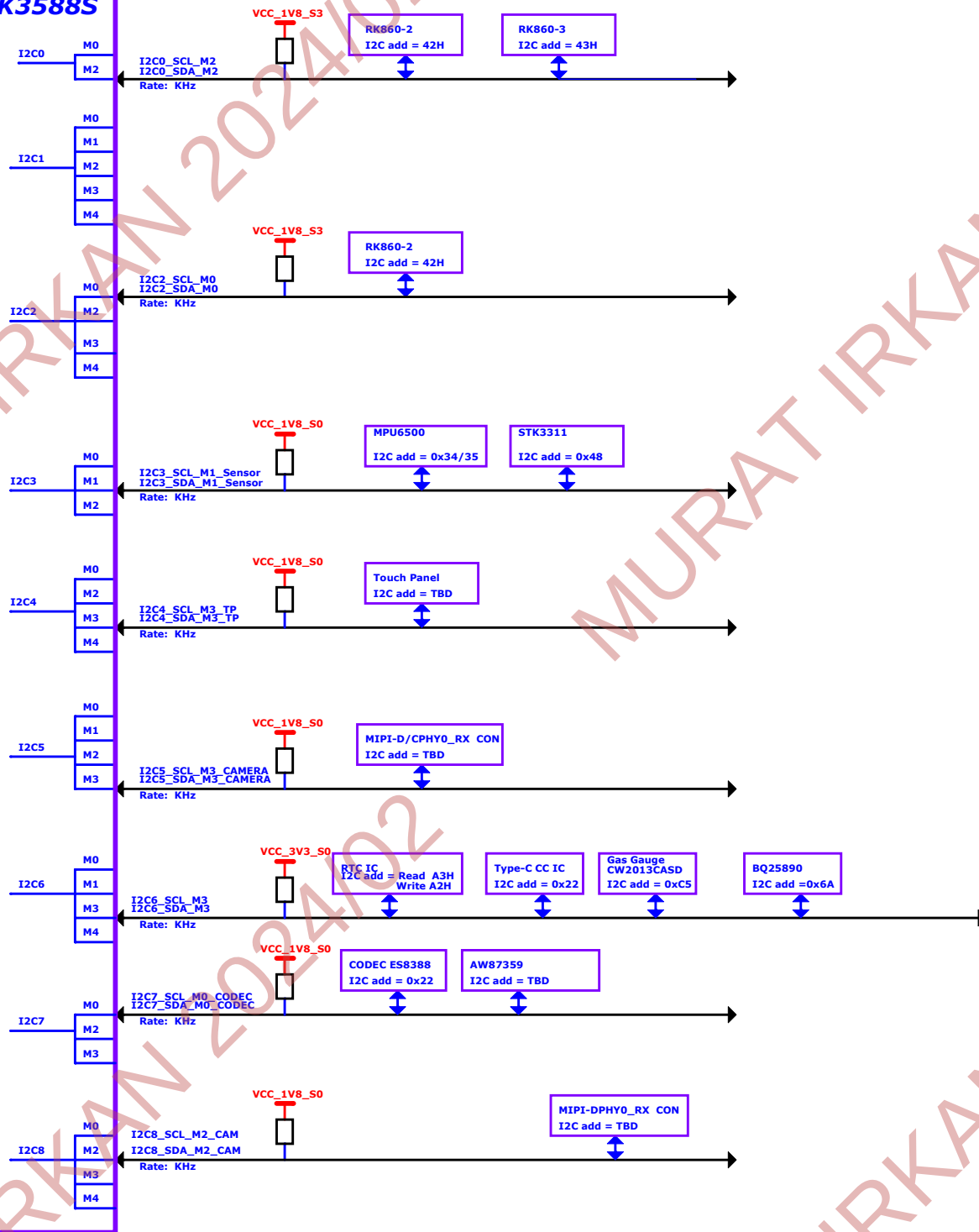
Reviewed by: <Checker>

Rev: V10

Sheet: 5 of 32

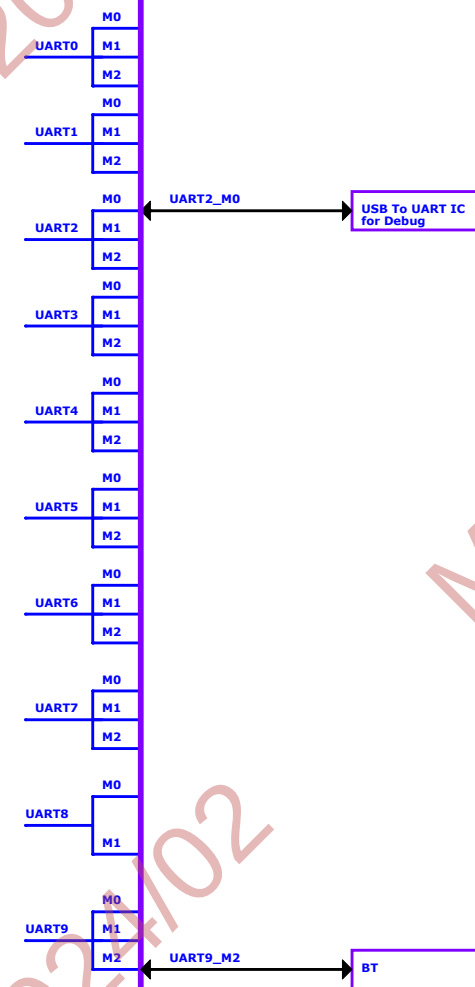
I2C MAP

RK3588S



UART MAP

RK3588S



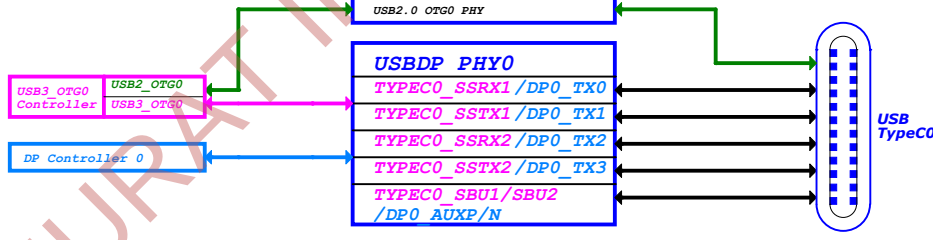
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USB Controller Configure Table

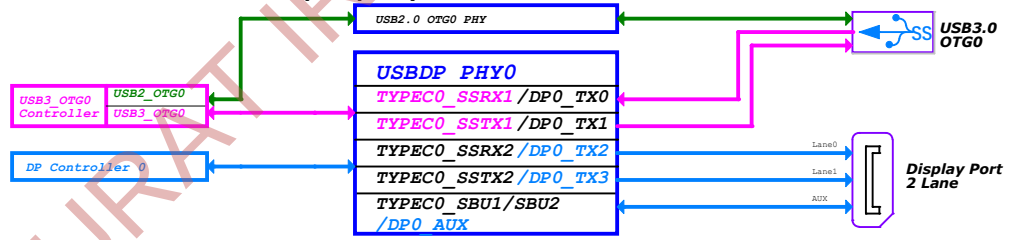
Controller Name	Pin Name	Type-C Function	DPxLane+USB20 OTG		USB30 OTG+DPx2Lane Function		USB20 OTG+DPx2Lane Function		USB20 OTG+DPx4Lane Function	
			OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
USB30 OTG0 Device or Host	TYPEC0_SBU1/DP0_AUX0	TYPEC0_SBU1	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0
	TYPEC0_SBU2/DP0_AUX0	TYPEC0_SBU2	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0	DP0_AUX0
	TYPEC0_SSRX1/DP0_TX0	TYPEC0_SSRX1P	DP0_TX0P	DP0_TX0P	TYPEC0_SSRX1P	DP0_TX0P	DP0_TX0P	DP0_TX0P	DP0_TX0P	DP0_TX0P
	TYPEC0_SSRX1N/DP0_TX0N	TYPEC0_SSRX1N	DP0_TX0N	DP0_TX0N	TYPEC0_SSRX1N	DP0_TX0N	DP0_TX0N	DP0_TX0N	DP0_TX0N	DP0_TX0N
USB30 OTG0 Device or Host	TYPEC0_SSTX1/DP0_TX1	TYPEC0_SSTX1P	DP0_TX1P	DP0_TX1P	TYPEC0_SSTX1P	DP0_TX1P	DP0_TX1P	DP0_TX1P	DP0_TX1P	DP0_TX1P
	TYPEC0_SSTX1N/DP0_TX1N	TYPEC0_SSTX1N	DP0_TX1N	DP0_TX1N	TYPEC0_SSTX1N	DP0_TX1N	DP0_TX1N	DP0_TX1N	DP0_TX1N	DP0_TX1N
	TYPEC0_SSRX2/DP0_TX2	TYPEC0_SSRX2P	DP0_TX2P	DP0_TX2P	TYPEC0_SSRX2P	DP0_TX2P	DP0_TX2P	DP0_TX2P	DP0_TX2P	DP0_TX2P
	TYPEC0_SSRX2N/DP0_TX2N	TYPEC0_SSRX2N	DP0_TX2N	DP0_TX2N	TYPEC0_SSRX2N	DP0_TX2N	DP0_TX2N	DP0_TX2N	DP0_TX2N	DP0_TX2N
USB30 OTG0 Device or Host	TYPEC0_SSTX2/DP0_TX3	TYPEC0_SSTX2P	DP0_TX3P	DP0_TX3P	TYPEC0_SSTX2P	DP0_TX3P	DP0_TX3P	DP0_TX3P	DP0_TX3P	DP0_TX3P
	TYPEC0_SSTX2N/DP0_TX3N	TYPEC0_SSTX2N	DP0_TX3N	DP0_TX3N	TYPEC0_SSTX2N	DP0_TX3N	DP0_TX3N	DP0_TX3N	DP0_TX3N	DP0_TX3N
	TYPEC0_SBU1/DP0_AUX1	TYPEC0_SBU1	DP0_AUX1	DP0_AUX1	TYPEC0_SBU1	DP0_AUX1	DP0_AUX1	DP0_AUX1	DP0_AUX1	DP0_AUX1
	TYPEC0_SBU2/DP0_AUX1	TYPEC0_SBU2	DP0_AUX1	DP0_AUX1	TYPEC0_SBU2	DP0_AUX1	DP0_AUX1	DP0_AUX1	DP0_AUX1	DP0_AUX1
USB30 OTG2 Device or Host	PCIE20_2_RXP/SATA30_2_RXP/USB30_2_RXP	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
	PCIE20_2_RXN/SATA30_2_RXN/USB30_2_RXN	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
	PCIE20_2_RXP/SATA30_2_RXP/USB30_2_RXP	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
	PCIE20_2_RXN/SATA30_2_RXN/USB30_2_RXN	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
USB20 HOST0	USB20_HOST0_DP	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
	USB20_HOST0_DP	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
USB20 HOST1	USB20_HOST1_DP	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
	USB20_HOST1_DP	TYPEC0_OTG2_DP	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2

Note:
DP Lane swap enable
0: Lane0/1/2/3 TxData mapping to Lane0/1/2/3 TXDP/N
1: Lane0/1/2/3 TxData mapping to Lane2/3/0/1 TXDP/N

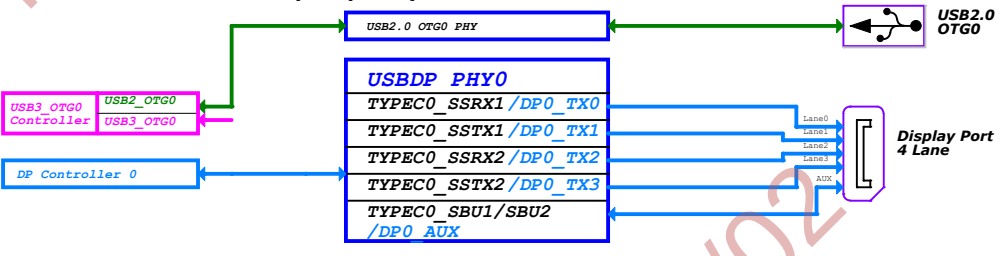
Config0: TypeC0 (With DP function)



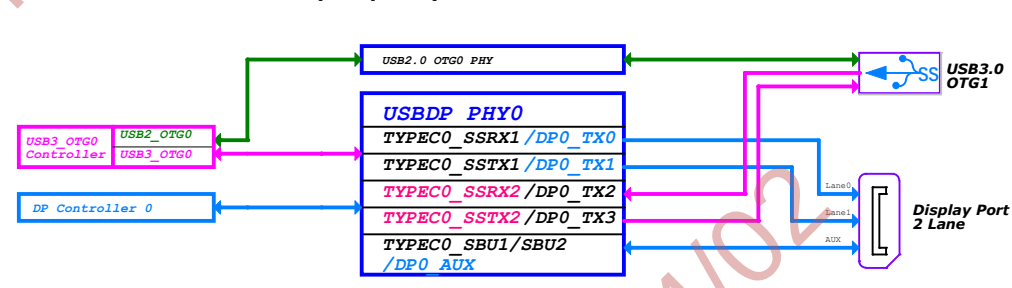
Config3:(Default) USB3.0 OTG0 + DP0 2Lane(Swap ON)



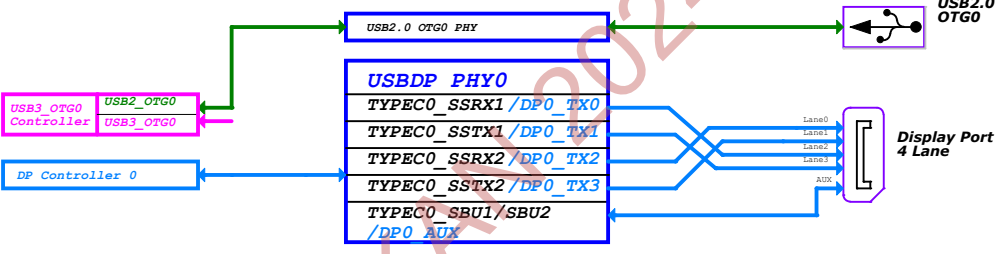
Config1: USB2.0 OTG0 + DP0 4Lane(Swap OFF)



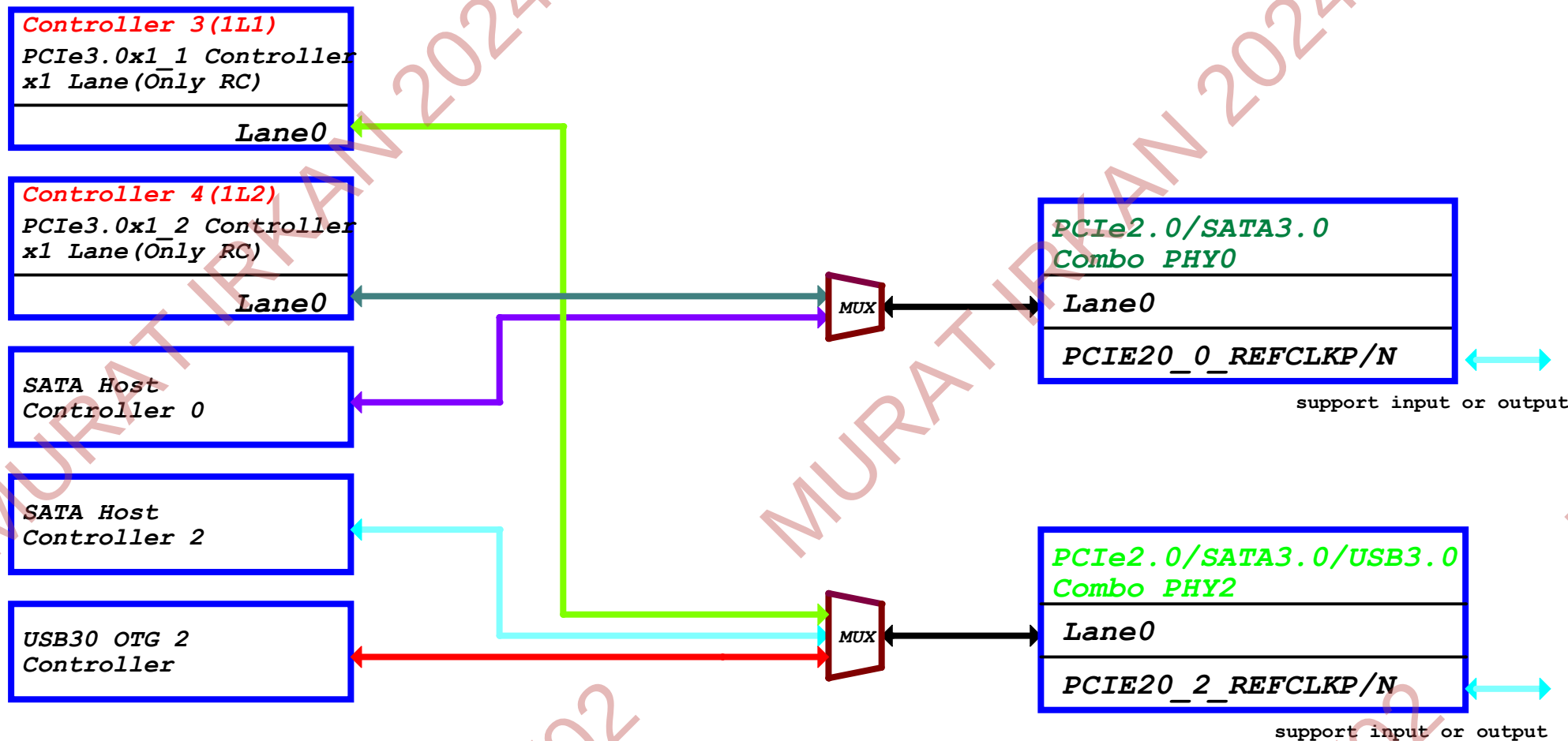
Config4: USB3.0 OTG0 + DP0 2Lane(Swap OFF)



Config2: USB2.0 OTG0 + DP0 4Lane(Swap ON)



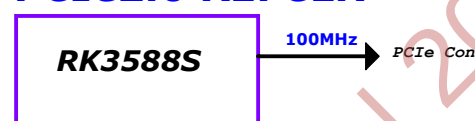
PCIe/SATA Connector Diagram



PCIe Controller Configure Table

Controller Name	Data & Clk Lane Configure		Control GPIO
	CLK LANE	DATA LANE	
PCIE20X1_1 RC	PCIE20_2_REFCLKP PCIE20_2_REFCLKN	PCIE20_2_TX PCIE20_2_RX	PCIE20X1_1_CLKREQ_M* PCIE20X1_1_WAKEN_M* PCIE20X1_1_PERSTN_M* PCIE20X1_1_BUTTON_RSTN
PCIE20X1_2 RC	PCIE20_0_REFCLKP PCIE20_0_REFCLKN	PCIE20_0_TX PCIE20_0_RX	PCIE20X1_2_CLKREQ_M* PCIE20X1_2_WAKEN_M* PCIE20X1_2_PERSTN_M* PCIE20X1_2_BUTTON_RSTN

PCIe2.0 REFCLK



Note:

PCIE20_*_REFCLKP/N is output or input gpio
M*=Mean to M0 or M1 or M2, it's the same source, Just multiplex to M0 or M1 or M2, Only use one at the same time.

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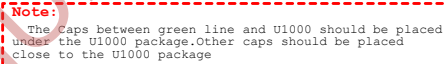
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Project: RK3588S_Demo

File: 08.PCIE Fun Map

Date: Friday, January 07, 2022 Rev: V10

Designed by: Joseph Reviewed by: <Checker> Sheet: 8 of 32



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Project:	RK3588S_Demo		
Date:	10.RK3588S_Power/GND		
File:	Friday, January 07, 2022	Rev:	V10
Designed by:	Joseph	Reviewed by:	<Checker>
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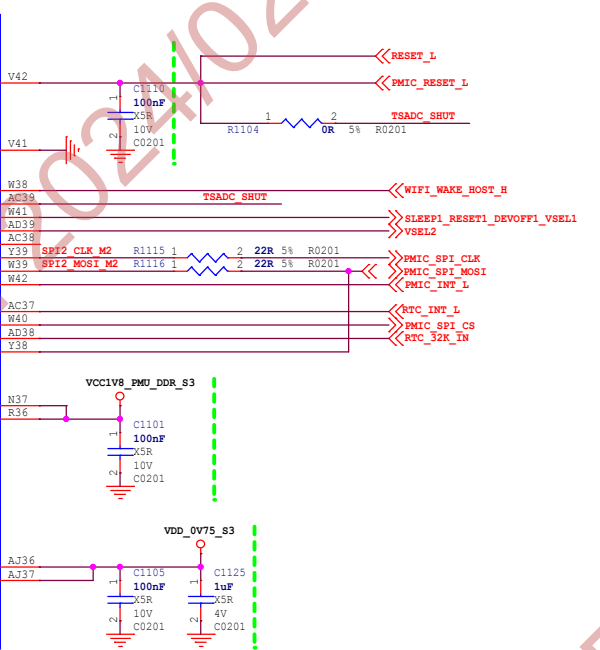
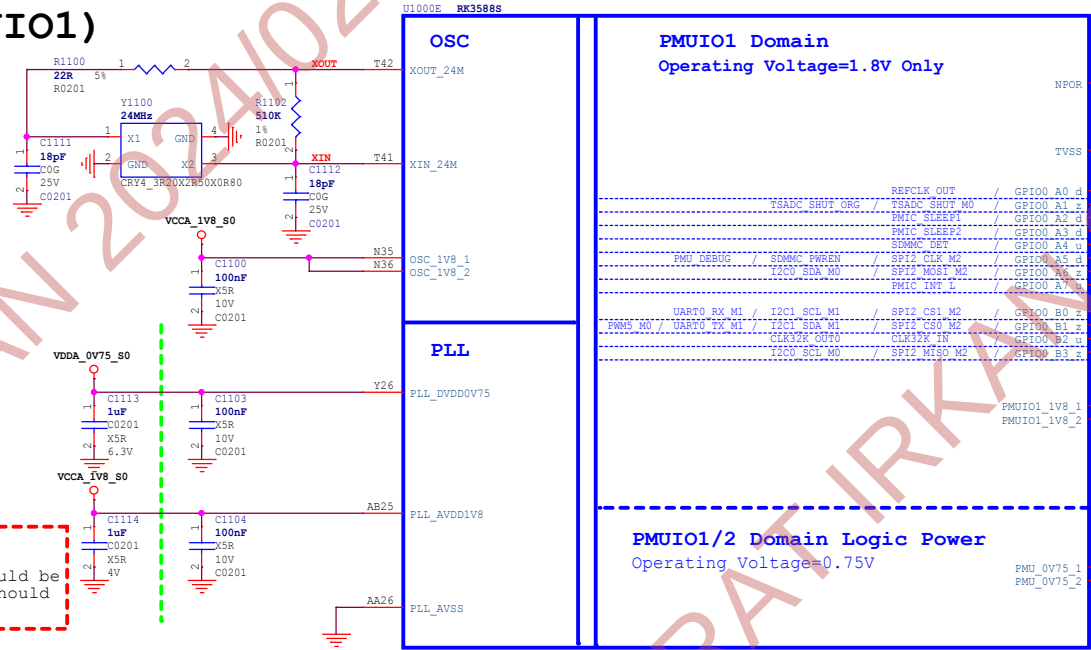
RK3588S (OSC/PLL/PMUIO1)

Note:
Adjusted the load capacitance according to the crystal specification

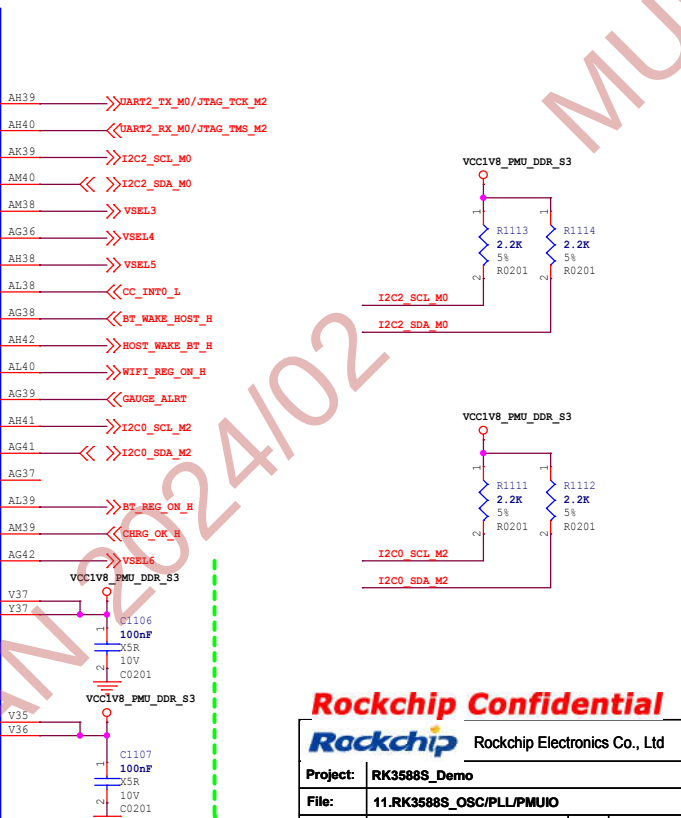
The CL is the load capacitance of the crystal that is recommended by the crystal vendors to obtain target clock frequency.

$CL = (CL1 * CL2 / (CL1 + CL2)) + PCB\ strays$
Total CL<=12pF

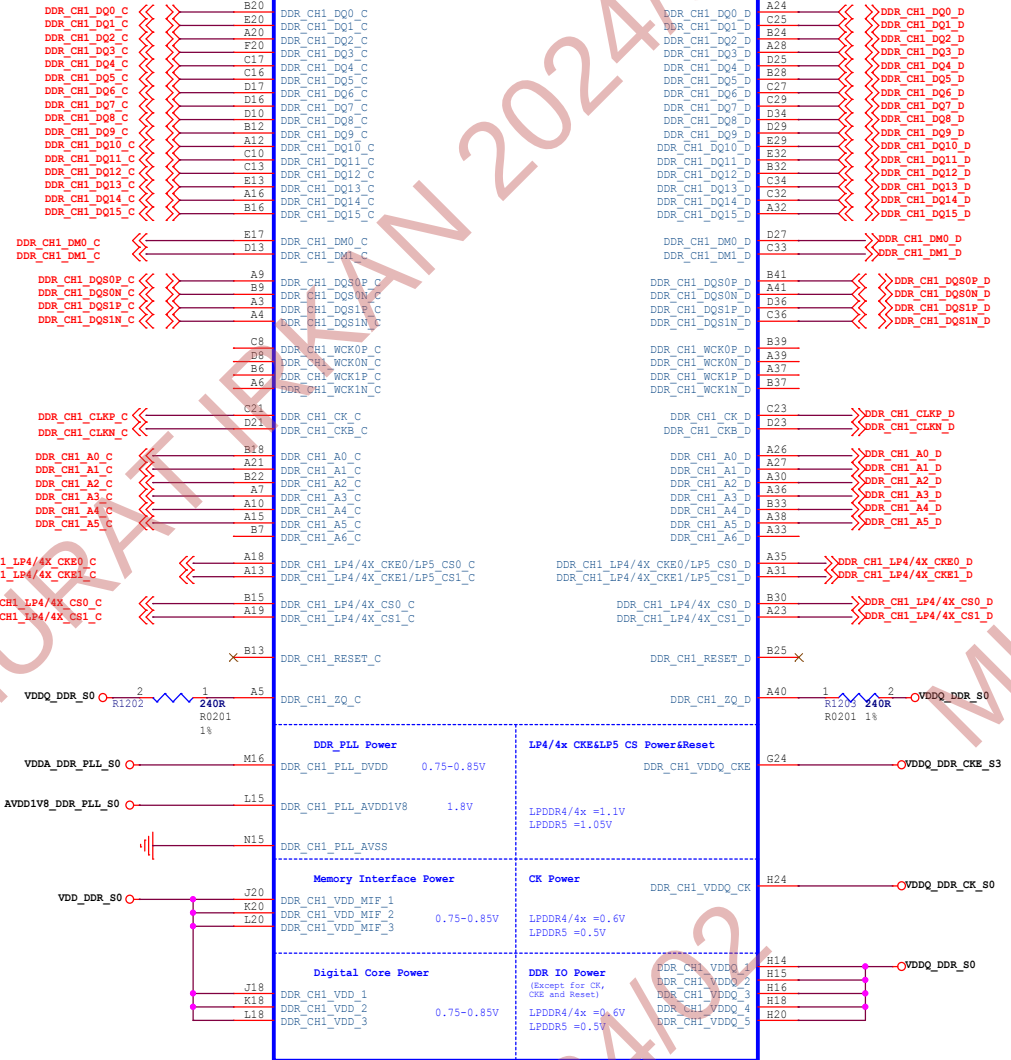
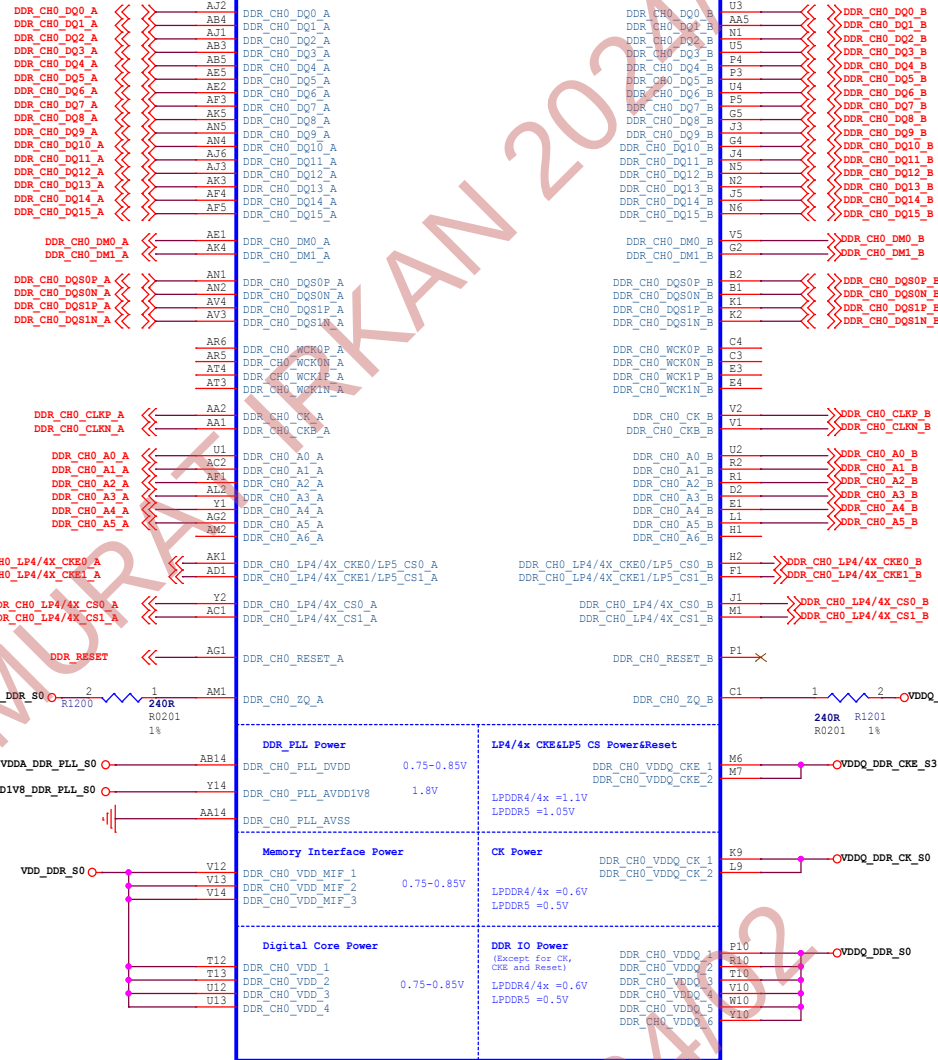
Note:
The Caps between green line and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package



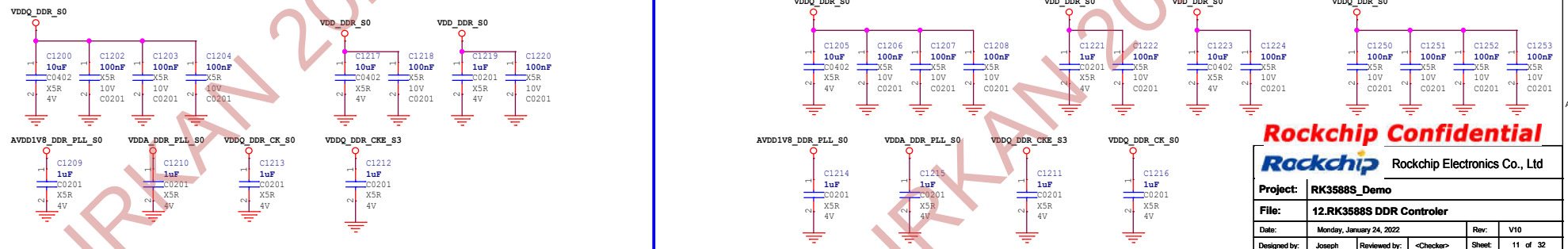
RK3588S (PMUIO2)



RK3588S (DDR PHY)



DDR FILTER



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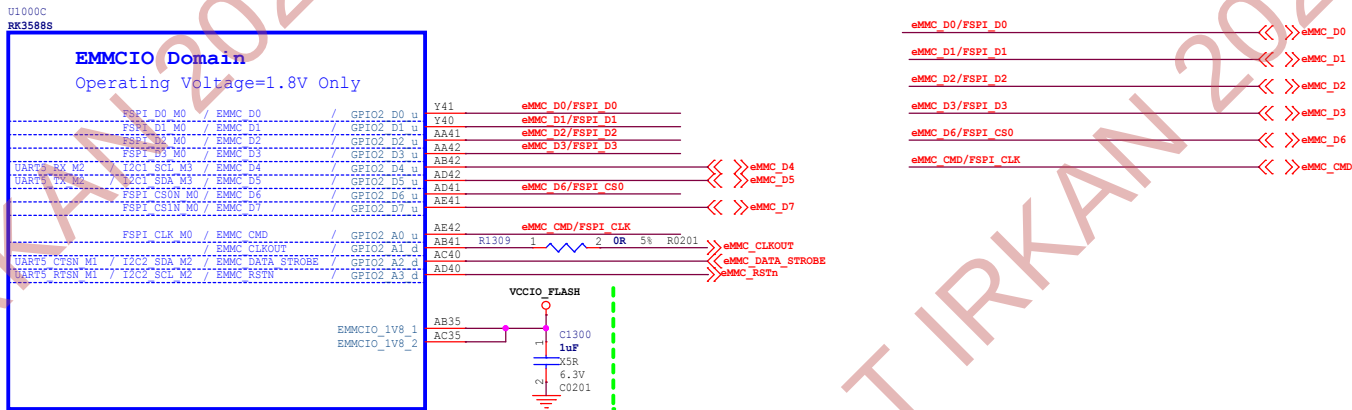
Project: RK3588S_Demo

File:	12.RK3588S DDR Controler
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Date:	Monday, January 24, 2022	Rev:	V10
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RK3588S (EMMCIO Domain)



RK3588S (VCCIO2 Domain)



Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

RK3588S (USB3.0/DP1.4)

USB30/DP1.4 Alt Mode Configuration

Option1	DP x4Lane	DP_TX_Lane0-3
Option2	TYPEC x4Lane	SSTX 1P/1N SSTX 2P/2N SSRX 1P/1N SSRX 2P/2N
Option3	USB30X2Lane+DPX2Lane	USB30:SSTX 1P/1N SSRX 1P/1N DP:Lane2 Lane3
Option4	USB30X2Lane+DPX2Lane	USB30:SSTX 2P/2N SSRX 2P/2N DP:Lane0 Lane1

DP Lane
Swap Off:
Lane0/1/2/3_TXdata mapping to Lane0/1/2/3_TXDP/N
Swap On:
Lane0/1/2/3_TXdata mapping to Lane2/3/0/1_TXDP/N

U1000L RK3588S

USB 3.0 OTG of TYPECO /DP1.4 ALT

USB:U3/Gen1
DP:RBR/HBR/HBR2/HBR3

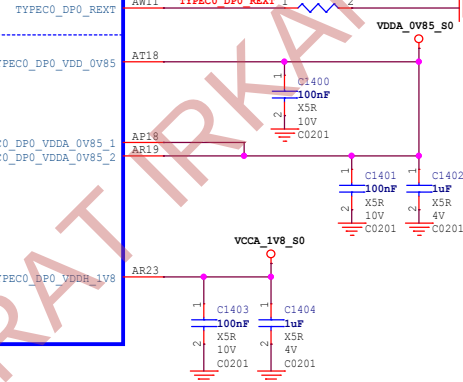
TYPECO_SBU1/DP0_AUXP BA8
TYPECO_SBU2/DP0_AUXN BB8
TYPECO_SSRX1P/DP0_TX0P BB10
TYPECO_SSRX1N/DP0_TX0N BA10
TYPECO_SSTX1P/DP0_TX1P BB11
TYPECO_SSTX1N/DP0_TX1N BA11

TYPECO_SSRX2P/DP0_TX2P BB13
TYPECO_SSRX2N/DP0_TX2N BA13
TYPECO_SSTX2P/DP0_TX3P BB14
TYPECO_SSTX2N/DP0_TX3N BA14

TYPEC&DP MUX Differential Pair:
DATE:95 Ohm +10%
For Typec

USB30 Differential Pair: DATE:90 Ohm +10%
For USB30
DP Differential Pair: DATE:100 Ohm +10%
For DP

POWER

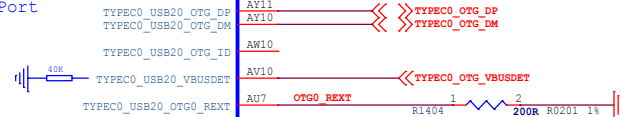


Do not delete!!!
If TYPECO is not used:
Signal:leave floating
REXT:8.2K ohm 1% resistor must
be connected externally
Power: Must supply power

RK3588S (USB2.0)

U1000K RK3588S

USB2.0 OTG of TYPECO HS/FS/LS Download Port



USB2.0 HOST0 HS/FS/LS

USB20_HOST0_DP AW6
USB20_HOST0_DM AV6
USB20_HOST0_REXT AU5

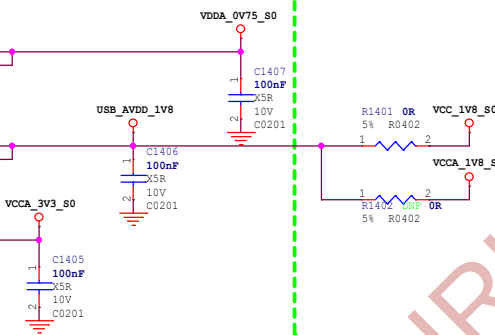
USB20 Differential Pair:
DATE:90 Ohm +10%

USB2.0 HOST1 HS/FS/LS

USB20_HOST1_DP AV7
USB20_HOST1_DM AW7
USB20_HOST1_REXT AU6

USB2.0 POWER

USB20_DVDD_0V75_1 AT11
USB20_DVDD_0V75_2 AT12
USB20_AVDD_1V8_1 AT13
USB20_AVDD_1V8_2 AT14
USB20_AVDD_3V3 AT10



Note:

The USB20_VBUSDET pin internal has a pull-down resistance(40K ohm) to ground,The resistance creates a voltage with the external series 24K ohm resistor.The VBUSDETpin voltage range <=3.3V.

Note:

TYPECO_USB20_OTG:
DP/DM:Must used for download
ID:According to demand,if not used,Leave floating
VBUSDET:Must provide
REXT:200ohm 1% resistor must be connected externally
Power: Must supply power

USB20_HOST0/USB20_HOST1:

If not used:
DP/DM:Leave floating
REXT:Leave floating

Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

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Project: RK3588S_Demo

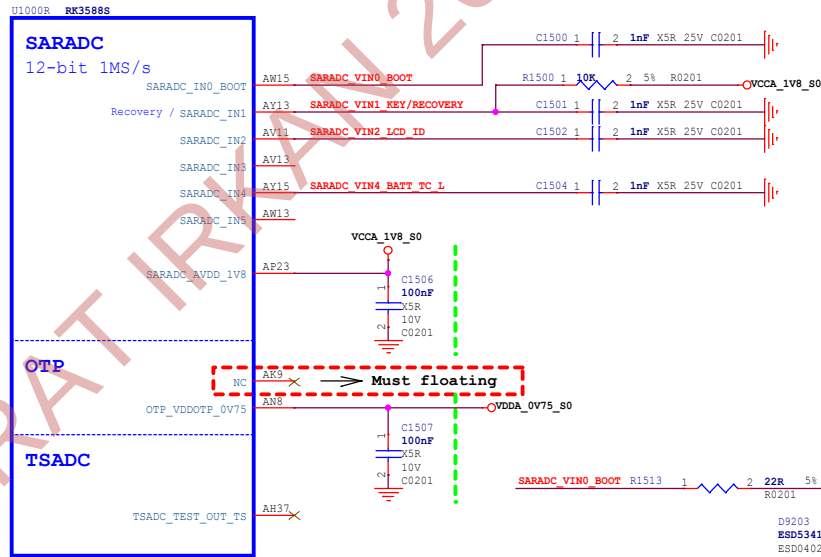
File: 14.RK3588S_USB20/USB30/DP_PHY

Date: Wednesday, February 23, 2022 Rev: V10

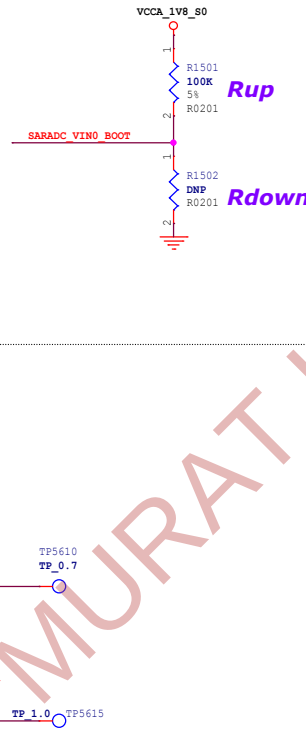
Designed by: Joseph Reviewed by: <Checker> Sheet: 13 of 32

RK3588S (SARADC/OTP/TSADC)

SARADC_VIN1_KEY/RECOVERY
 SARADC_VIN2_LCD_ID
 SARADC_VIN4_BATT_TC_L

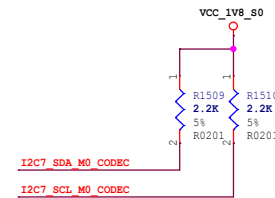
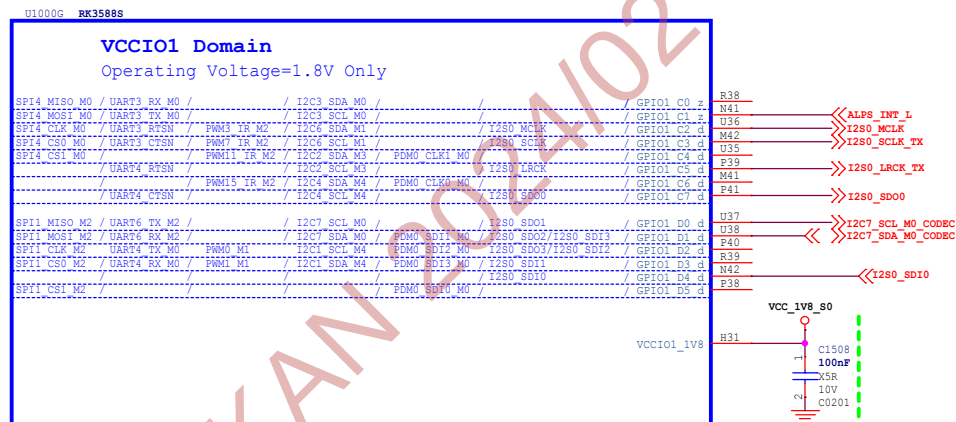


BOOT MODE CONFIG



Item	Rup	Rdown	ADC	BOOT MODE(saradc_in5)
LEVEL1	DNP	100K	0	USB (Maskrom mode)
LEVEL2	100K	20K	682	SD Card-USB
LEVEL3	100K	51K	1365	EMMC-USB
LEVEL4	100K	100K	2047	FSPI M0-USB
LEVEL5	100K	200K	2730	FSPI M1-USB
LEVEL6	100K	499K	3412	FSPI M2-USB
LEVEL7	100K	DNP	4095	FSPI M2-FSPI M0-EMMC -SD Card-USB

RK3588S (VCCIO1 Domain)



Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

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Project:	RK3588S_Demo		
File:	15.RK3588S_SARADC/1.8V GPIO		
Date:	Friday, January 14, 2022	Rev:	V10
Designed by:	Joseph	Reviewed by:	<Checker>
Sheet:	14	of	32

U1000N RK3588S





◎ 俗文化語彙 · 一



RK3588S (HDMI2.1 TX/eDP1.3 TX)

Note:

The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.

eDP TX
100 Ohm $\pm 10\%$

HDMI TX
100 Ohm $\pm 10\%$

U1000Q

HDMI TX/eDP1.3 MUX Port0

HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps

HDMI_TX0_D0P/EDP_TX0_D0P BB4
HDMI_TX0_D0N/EDP_TX0_D0N BA4

HDMI_TX0_D1P/EDP_TX0_D1P BA5
HDMI_TX0_D1N/EDP_TX0_D1N BB5

HDMI_TX0_D2P/EDP_TX0_D2P BB7
HDMI_TX0_D2N/EDP_TX0_D2N BA7

HDMI_TX0_D3P/EDP_TX0_D3P BA2
HDMI_TX0_D3N/EDP_TX0_D3N BB2

HDMI_TX0_SBDP/EDP_TX0_AUXP BA1
HDMI_TX0_SBDN/EDP_TX0_AUXN AY1

HDMI/eDP_TX0_REXT

AY3 HDMI/eDP_TX0_REXT R1708 1 2 8.2K 1% R0201 VDDA_0V75_S0

POWER

HDMI/EDP_TX0_VDD_0V75_1
HDMI/EDP_TX0_VDD_0V75_2

AM13
AN12
C1710 100nF X5R 10V C0201
C1711 4.7uF X5R 6.3V C0402

HDMI/EDP_TX0_AVDD_0V75

AN10
C1712 100nF X5R 10V C0201
C1725 1uF X5R 4V C0201

HDMI/EDP_TX0_VDD_IO_1V8
HDMI/EDP_TX0_VDD_CMN_1V8

AK12
AL14
C1713 100nF X5R 10V C0201
C1714 4.7uF X5R 6.3V C0402

VCCA_1V8_S0

Note:

If not used:
Signal: leave floating
Power: Floating or tie to VSS

Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

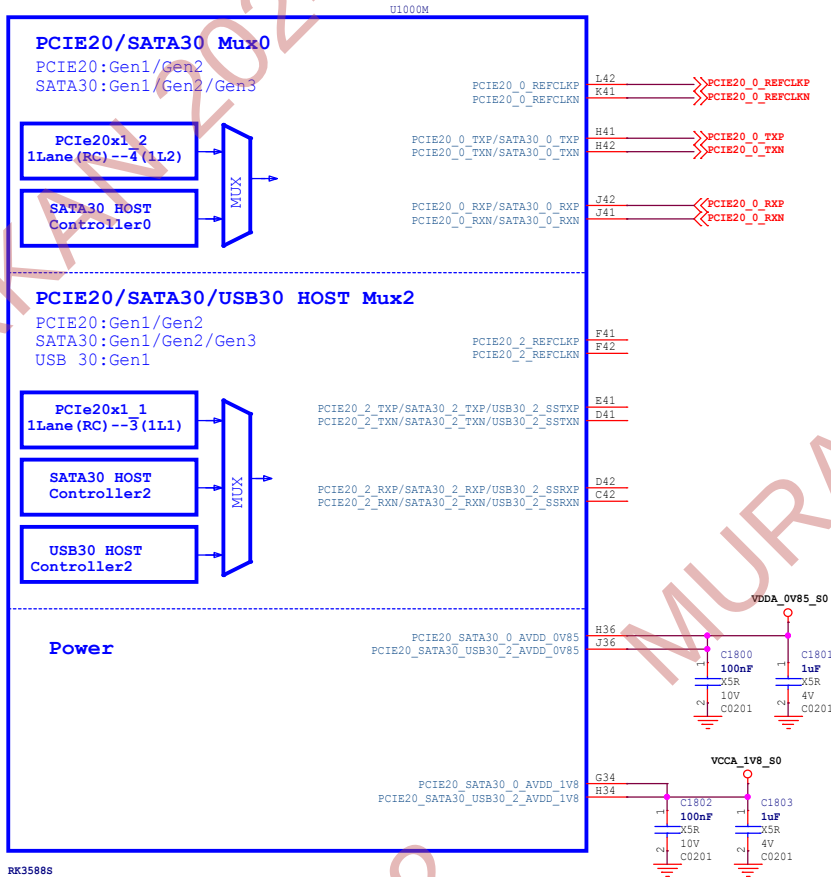
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Project:	RK3588S_Demo				
File:	17.RK3588S_HDMI/eDP Interface				
Date:	Friday, January 07, 2022			Rev:	V10
Designed by:	Joseph	Reviewed by:	<Checker>	Sheet:	16 of 32

RK3588S (PCIE20/SATA30/USB30)



PCIE2.0 PHY

Controller Name	Data & Clk Lane Configure		Control GPIO
	CLK LANE	DATA LANE	
PCIE20X1_1 RC	PCIE20_2_REFCLKP PCIE20_2_REFCLKN	PCIE20_2_TX PCIE20_2_RX	PCIE20X1_1_CLKREQ M* PCIE20X1_1_WAKEN M* PCIE20X1_1_PERSTN M* PCIE20X1_1_BUTTON_RSTN
PCIE20X1_2 RC	PCIE20_0_REFCLKP PCIE20_0_REFCLKN	PCIE20_0_TX PCIE20_0_RX	PCIE20X1_2_CLKREQ M* PCIE20X1_2_WAKEN M* PCIE20X1_2_PERSTN M* PCIE20X1_2_BUTTON_RSTN

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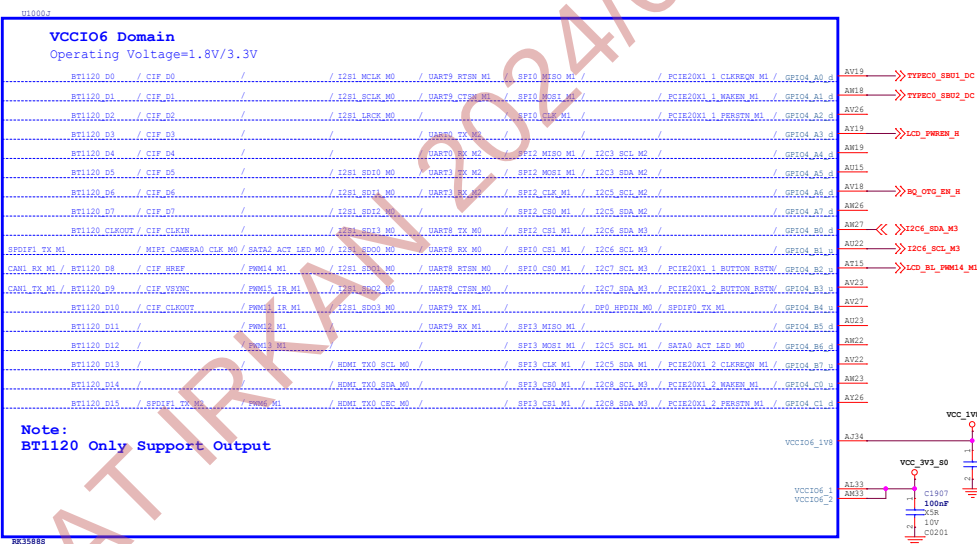
Project: RK3588S_Demo

File: 18.RK3588S_PCIE2/SATA3/USB3_PHY

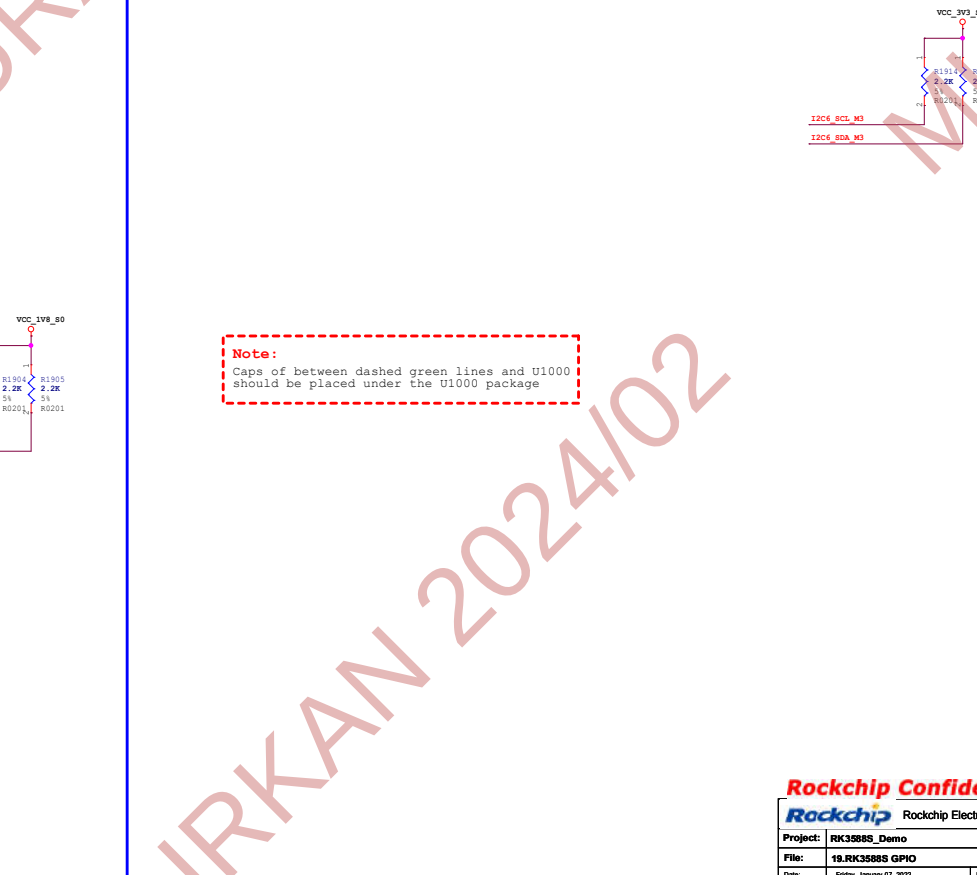
Date: Friday, January 07, 2022 Rev: V10

Designed by: Joseph Reviewed by: <Checker> Sheet: 17 of 32

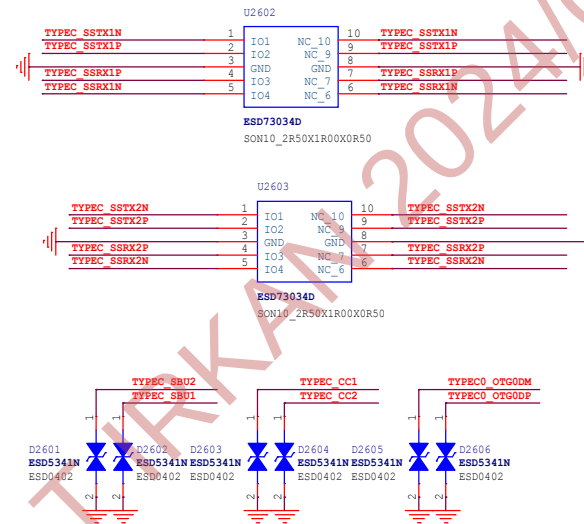
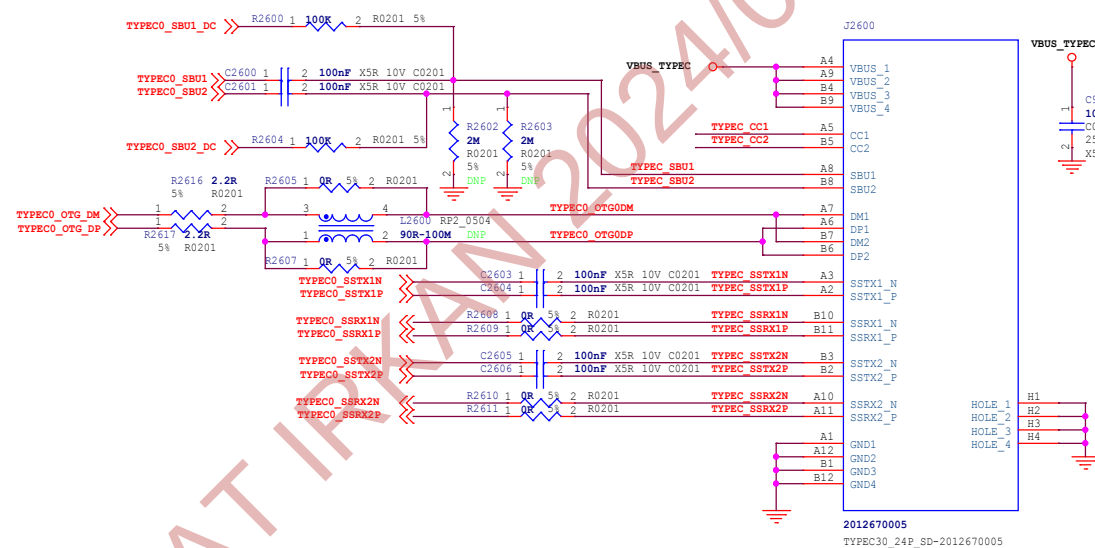
RK3588S (VCCI06 Domain)



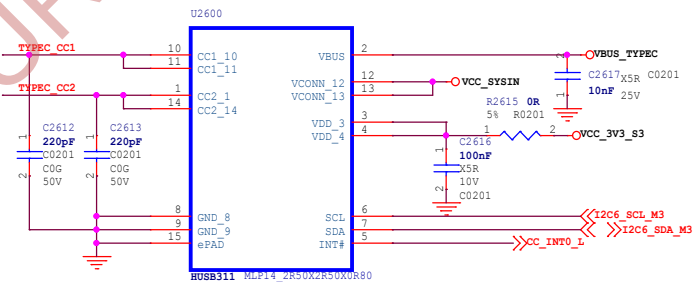
Note:
BT1120 Only Support Output



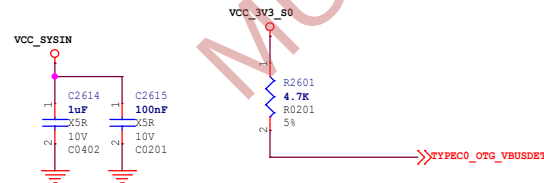
USB Type-C Port



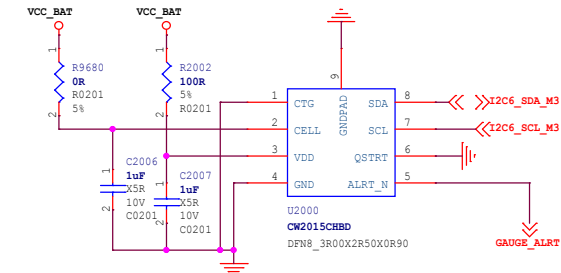
USB Type-C CC CTRL



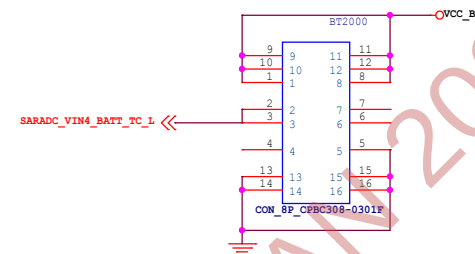
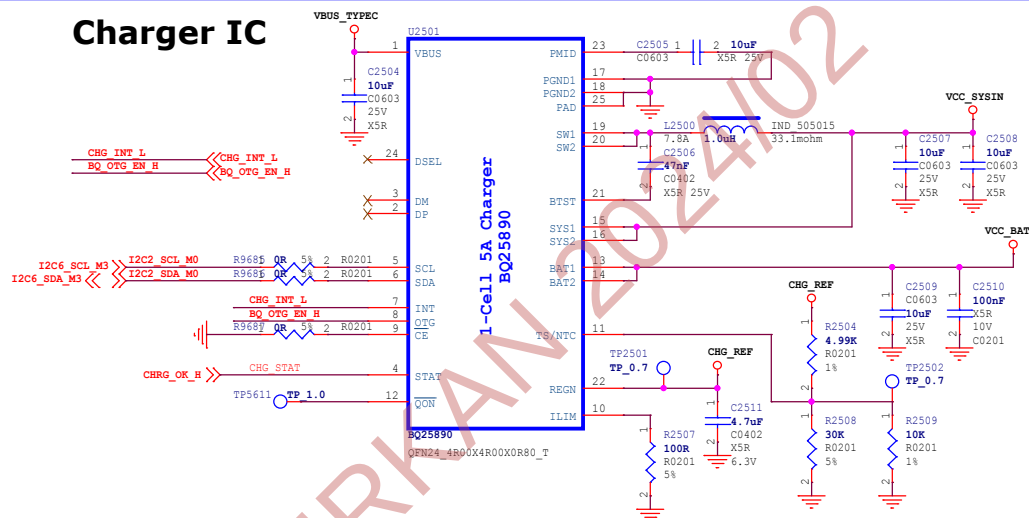
USB Detection



Gas Gauge



Charger IC



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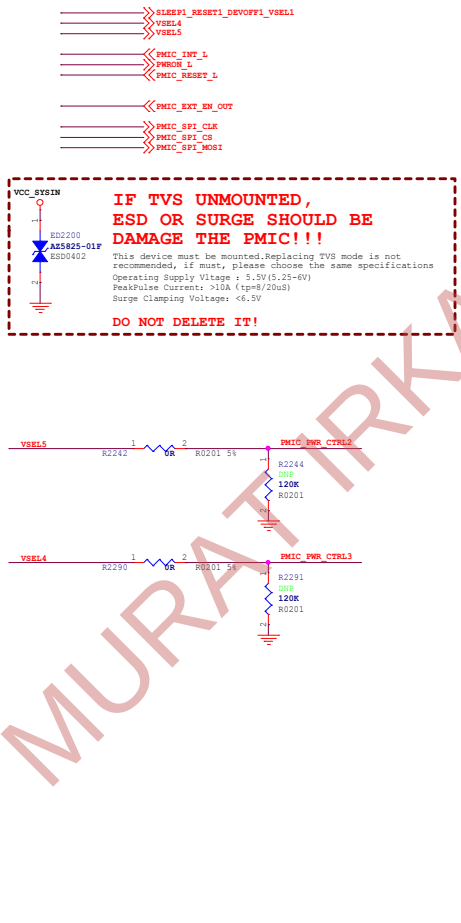
Project:	RK3588S_Demo
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File:	20.Power_1Cell_QC
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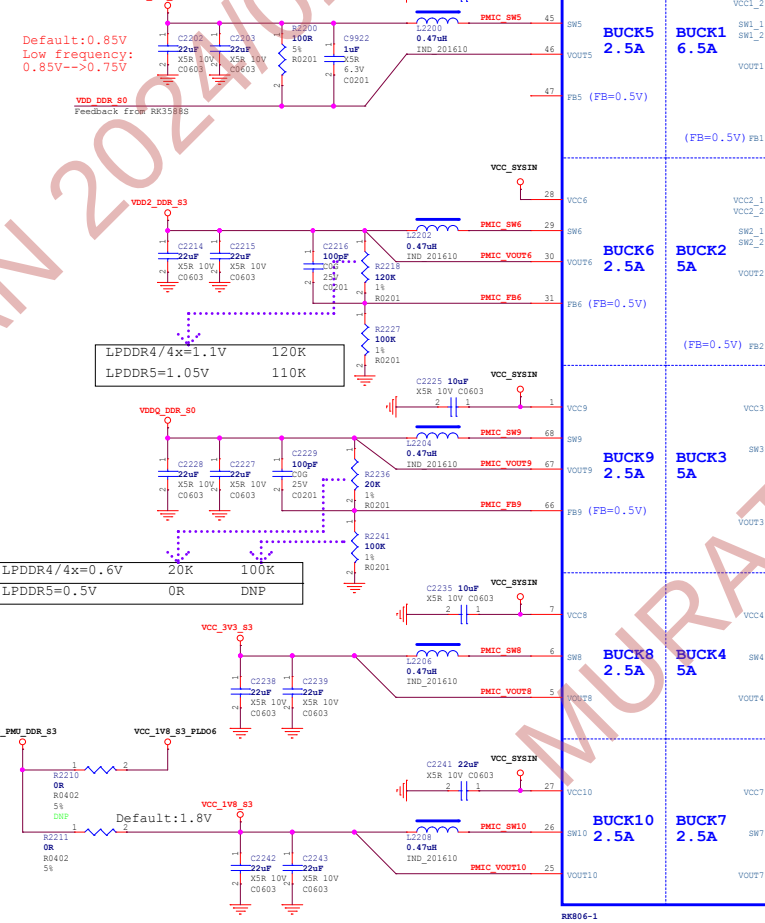
Date:	Friday, January 07, 2022	Rev:	V10
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Designed by:	Joseph	Reviewed by:	<Checker>	Sheet:	19 of 32
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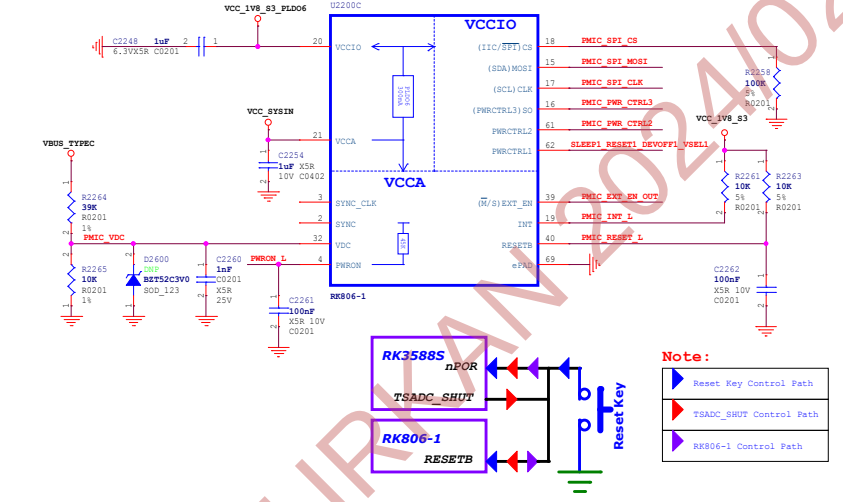
PMIC1 RK806-1



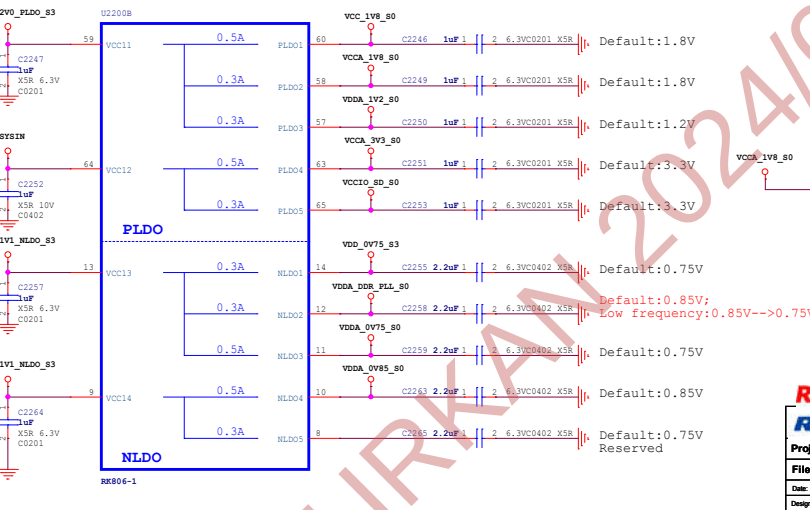
PMIC RK806-1 BUCK



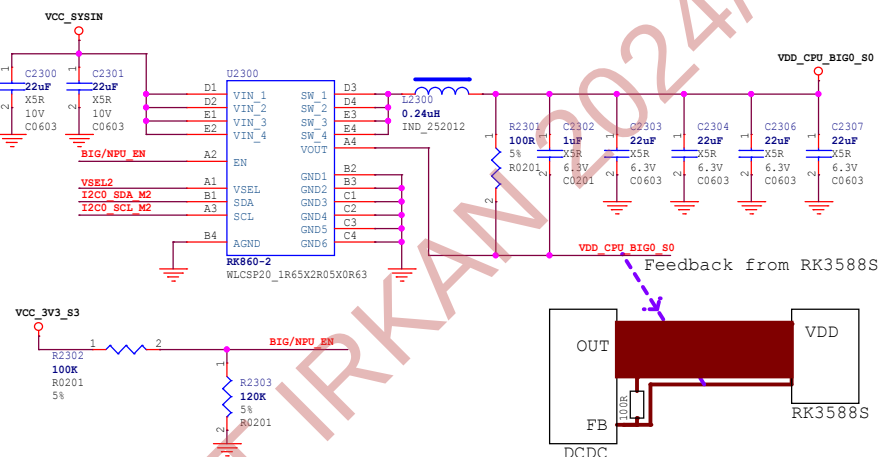
PMIC RK806-1 Management



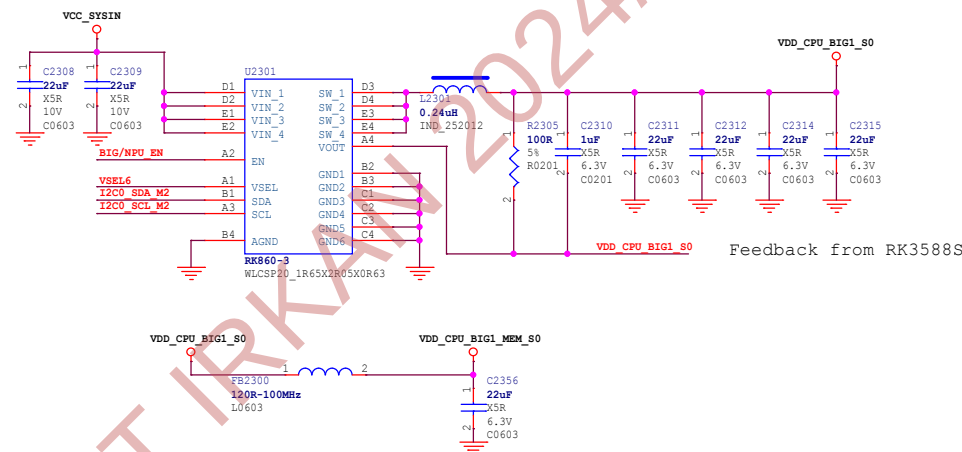
PMIC RK806-1 LDO



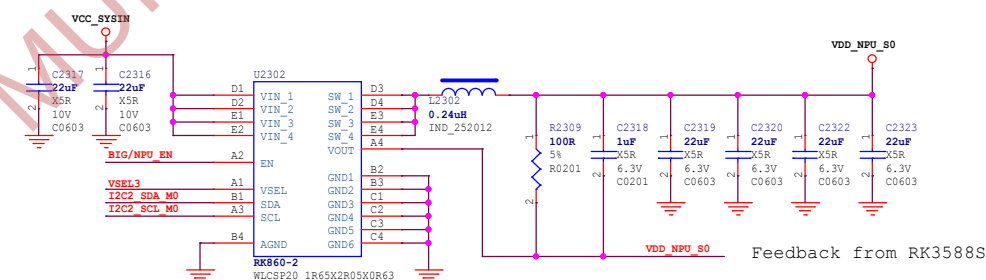
VDD_CPU_BIG0



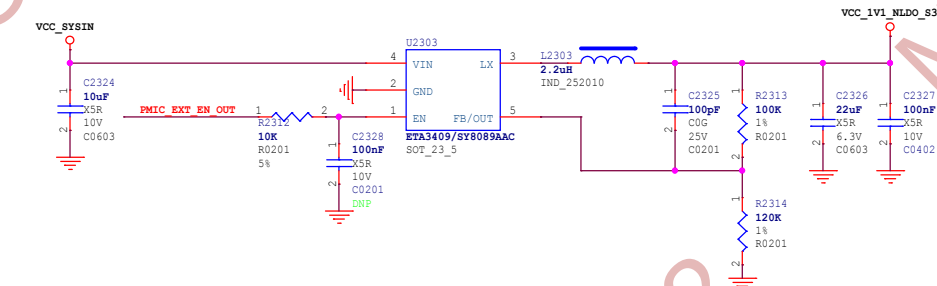
VDD_CPU_BIG1



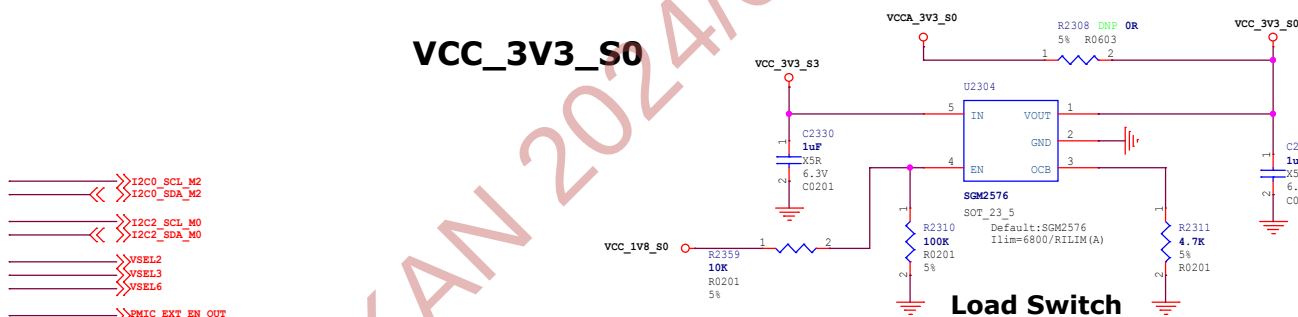
VDD_NPU



VCC_1V1_NLDO



VCC_3V3_S0

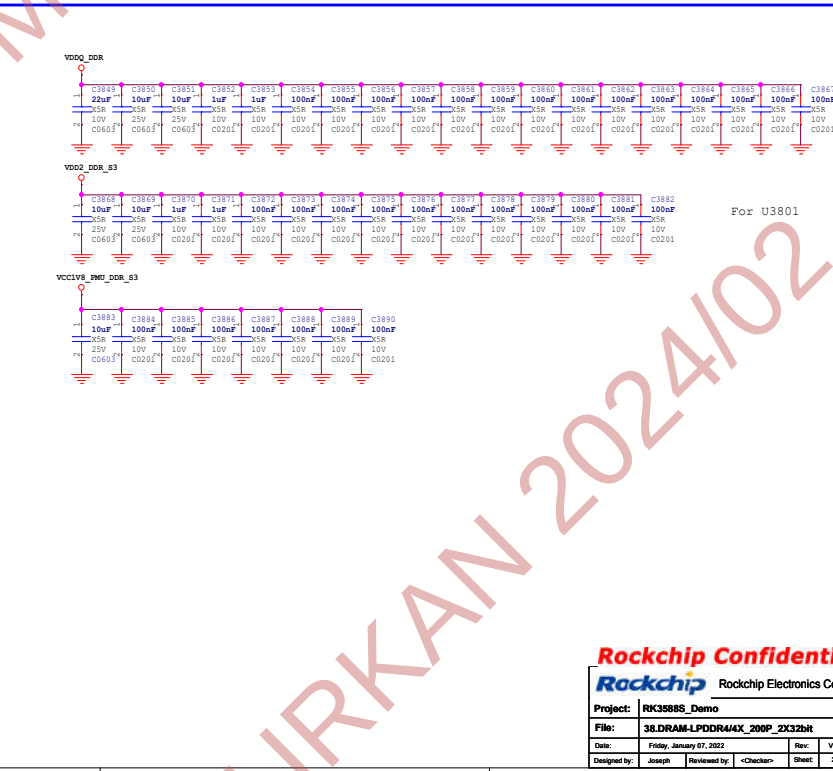
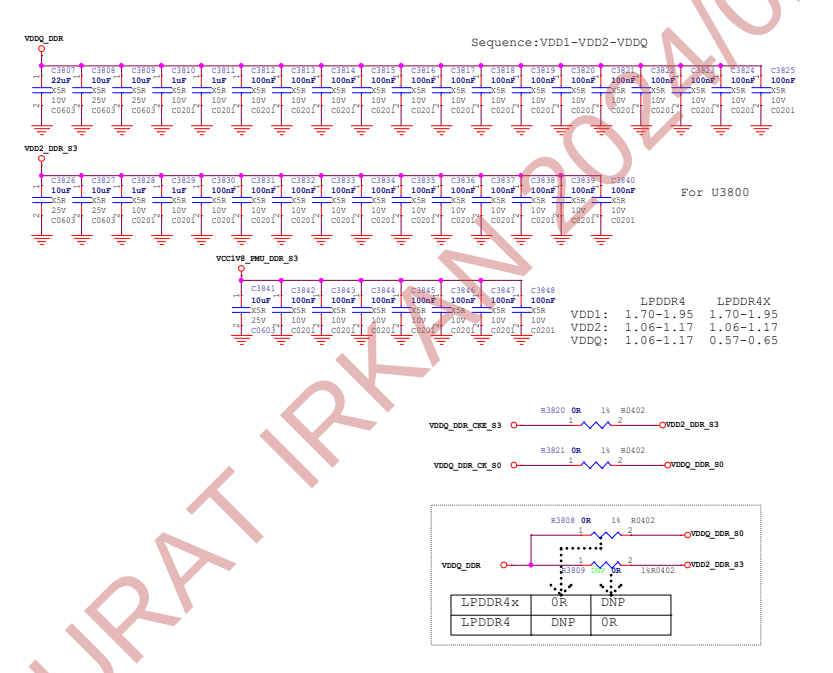


Load Switch

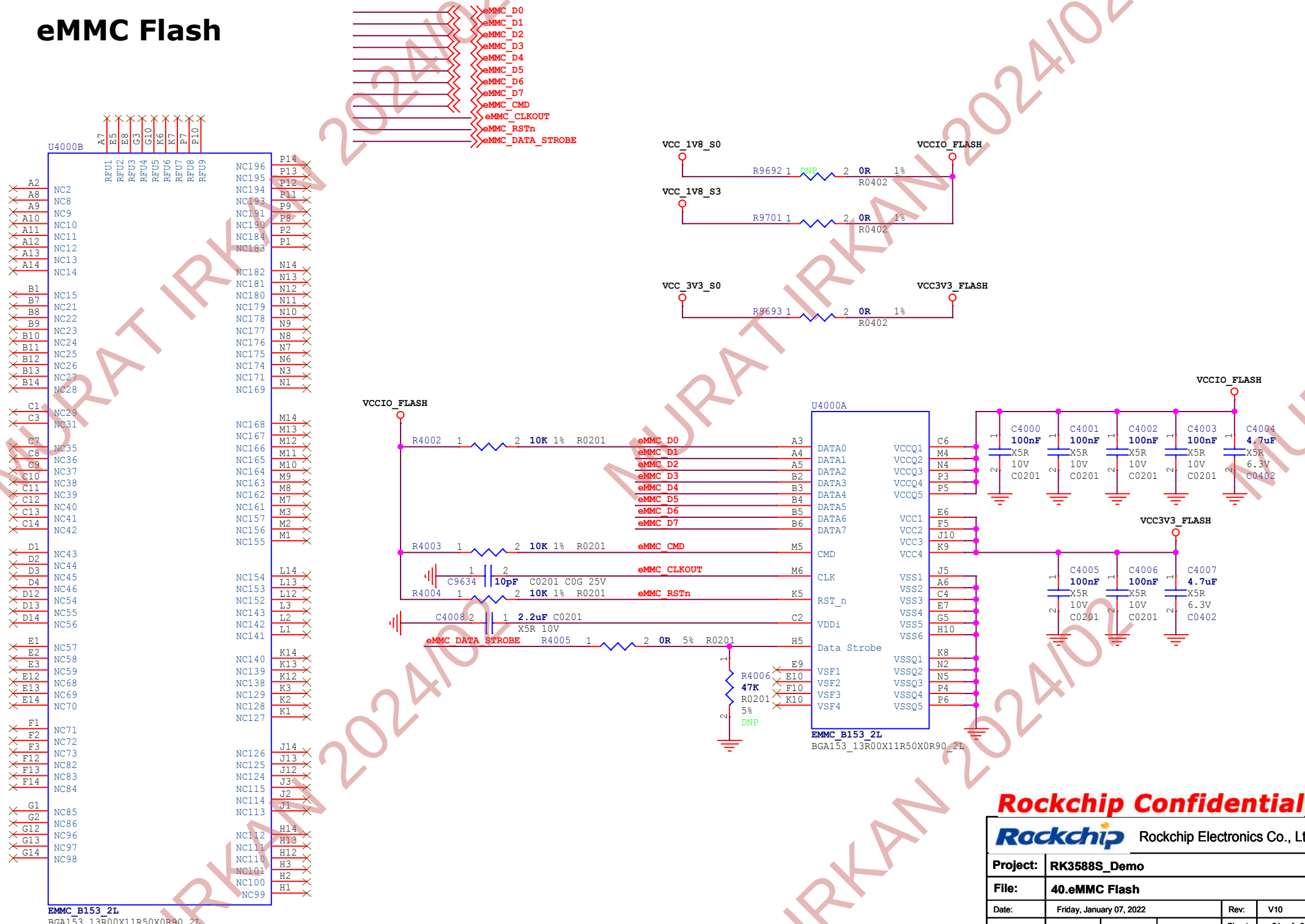
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Rackchip Rockchip Electronics Co., Ltd

Project:	RK3588S_Demo		
File:	23.Power_Ext Discrete		
Date:	Wednesday, February 23, 2022	Rev:	V10
Designed by:	Joseph	Reviewed by:	<Checker>
Sheet:	21	of	32

U3800A	
CH A	CH B



eMMC Flash



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Project: RK3588S_Demo

File: 40.eMMC Flash

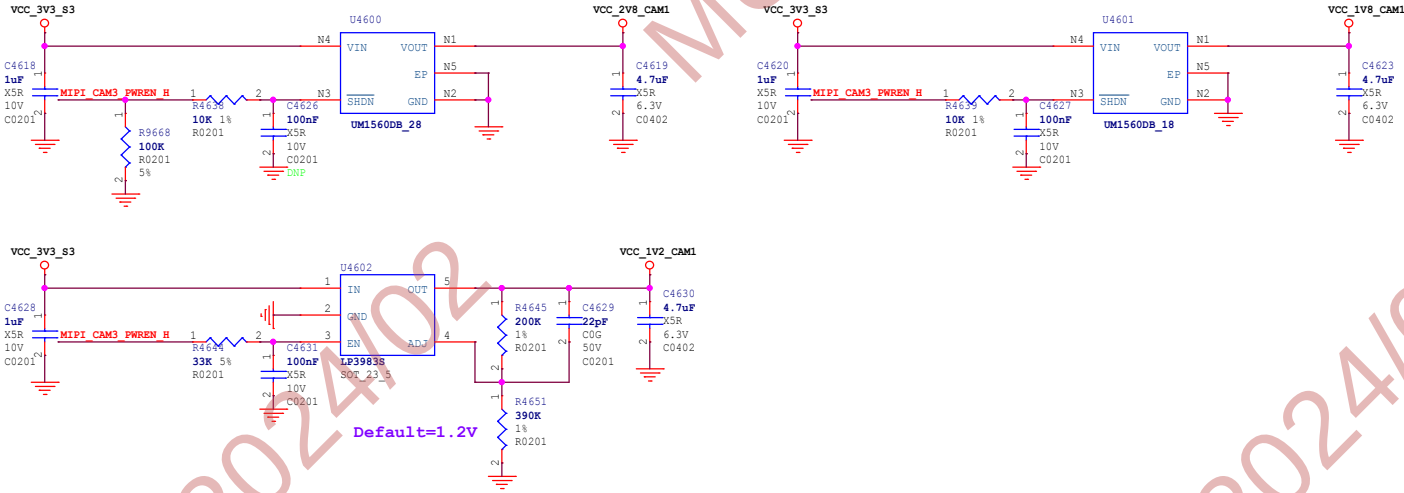
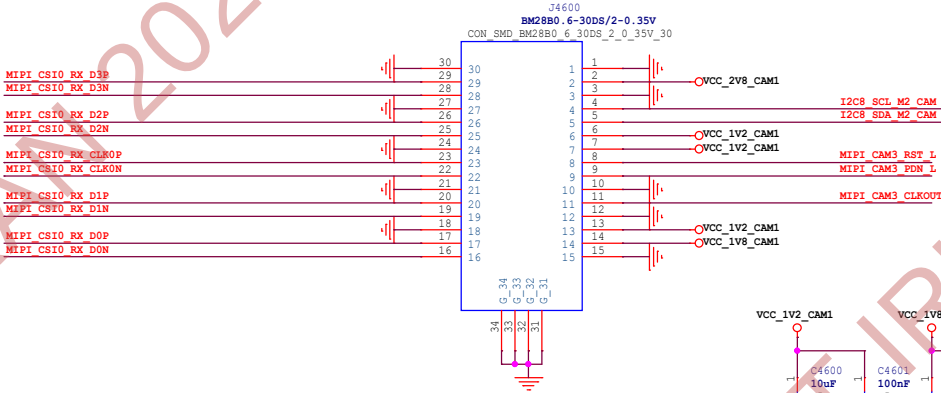
Date: Friday, January 07, 2022 **Rev:** V10

Designed by: Joseph **Reviewed by:** <Checker> **Sheet:** 24 of 32

MIPI-CSIO_RX

- MIPI_CSIO_RX_D0P
- MIPI_CSIO_RX_D0N
- MIPI_CSIO_RX_D1P
- MIPI_CSIO_RX_D1N
- MIPI_CSIO_RX_D2P
- MIPI_CSIO_RX_D2N
- MIPI_CSIO_RX_D3P
- MIPI_CSIO_RX_D3N
- MIPI_CSIO_RX_CLK0P
- MIPI_CSIO_RX_CLK0N
- MIPI_CAM3_CLKOUT
- MIPI_CAM3_RST_L
- MIPI_CAM3_PWREN_H
- MIPI_CAM3_PDN_L
- I2C8_SCL_M2_CAM
- I2C8_SDA_M2_CAM

前摄: OV16A10-GA5A-Z



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Project:	RK3588S_Demo			
File:	46.VI-Camera MIPI CSIO-RX			
Date:	Friday, January 07, 2022	Rev:	V10	
Designed by:	Joseph	Reviewed by:	<Checker>	Sheet: 25 of 32

VI-Camera DPHY_RX

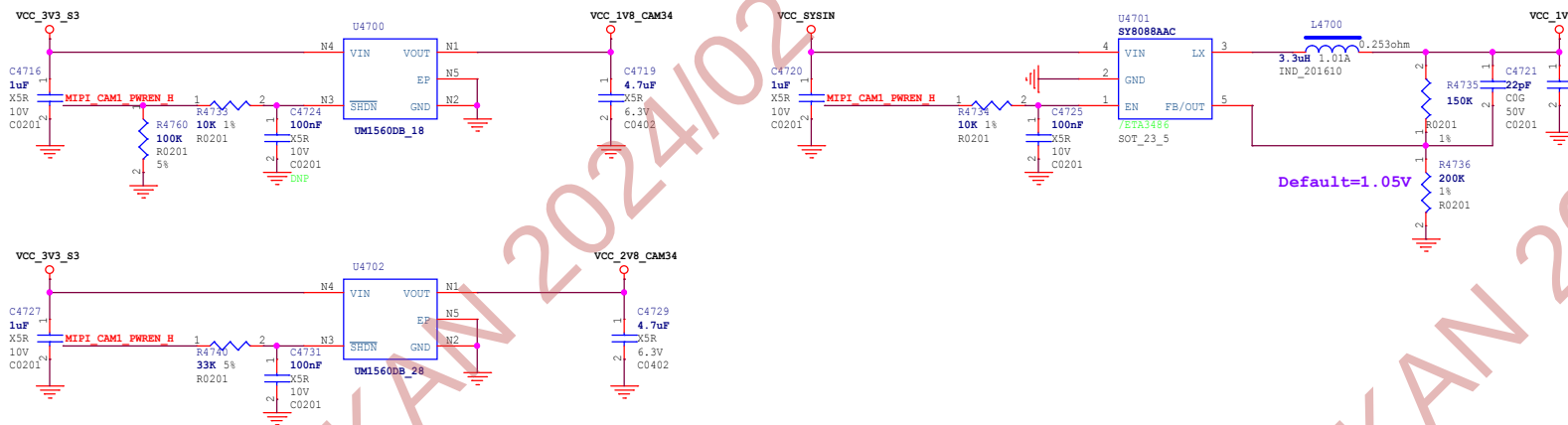
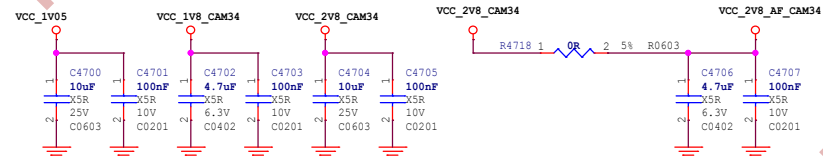
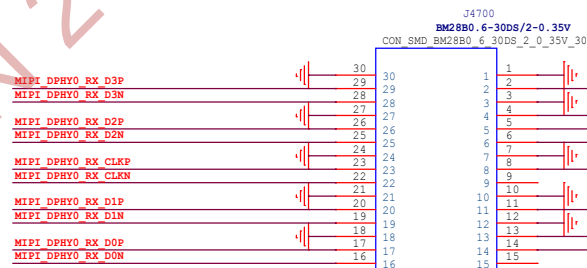
后摄: S5K3L6XX03-FGX9

MIIPI_DPHY0_RX_CLKP
MIIPI_DPHY0_RX_CLKN
MIIPI_DPHY0_RX_D0P
MIIPI_DPHY0_RX_D0N
MIIPI_DPHY0_RX_D1P
MIIPI_DPHY0_RX_D1N
MIIPI_DPHY0_RX_D2P
MIIPI_DPHY0_RX_D2N
MIIPI_DPHY0_RX_D3P
MIIPI_DPHY0_RX_D3N

MIIPI_CAM1_CLKOUT

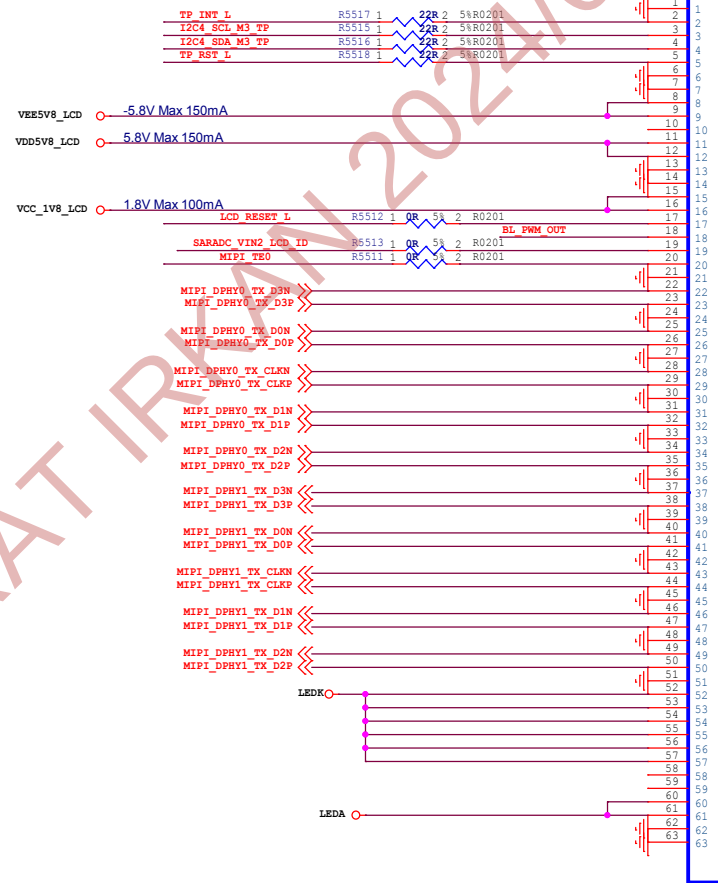
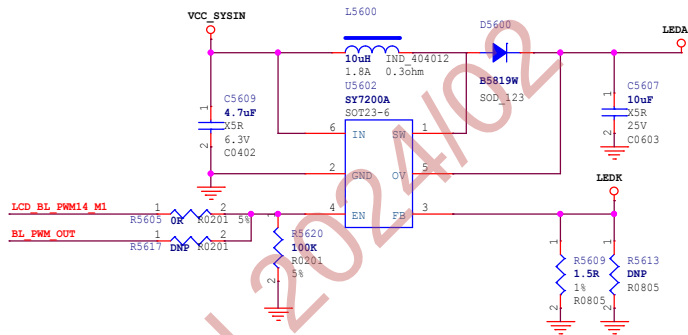
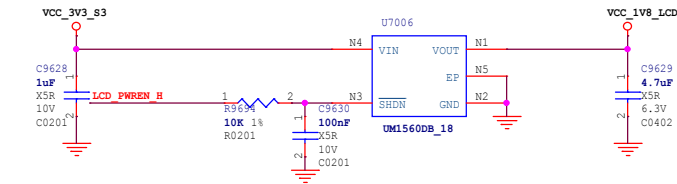
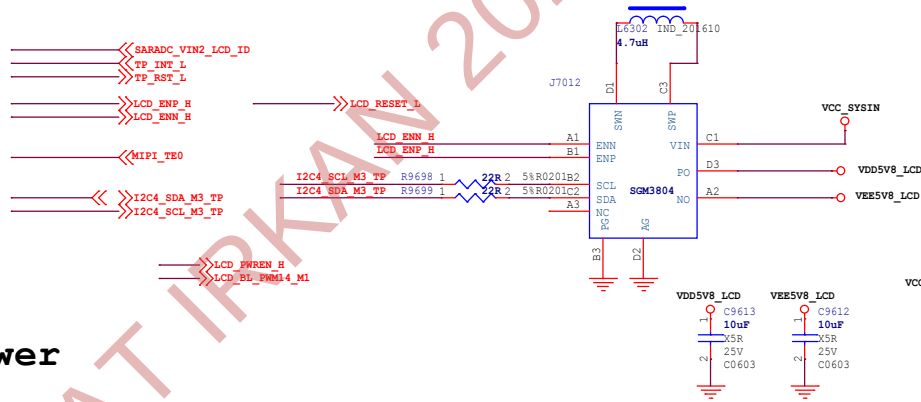
MIIPI_CAM1_PWREN_H
MIIPI_CAM1_RST_L

I2C5_SCL_M3_CAM
I2C5_SDA_M3_CAM



MIPI DPHY TX

Power

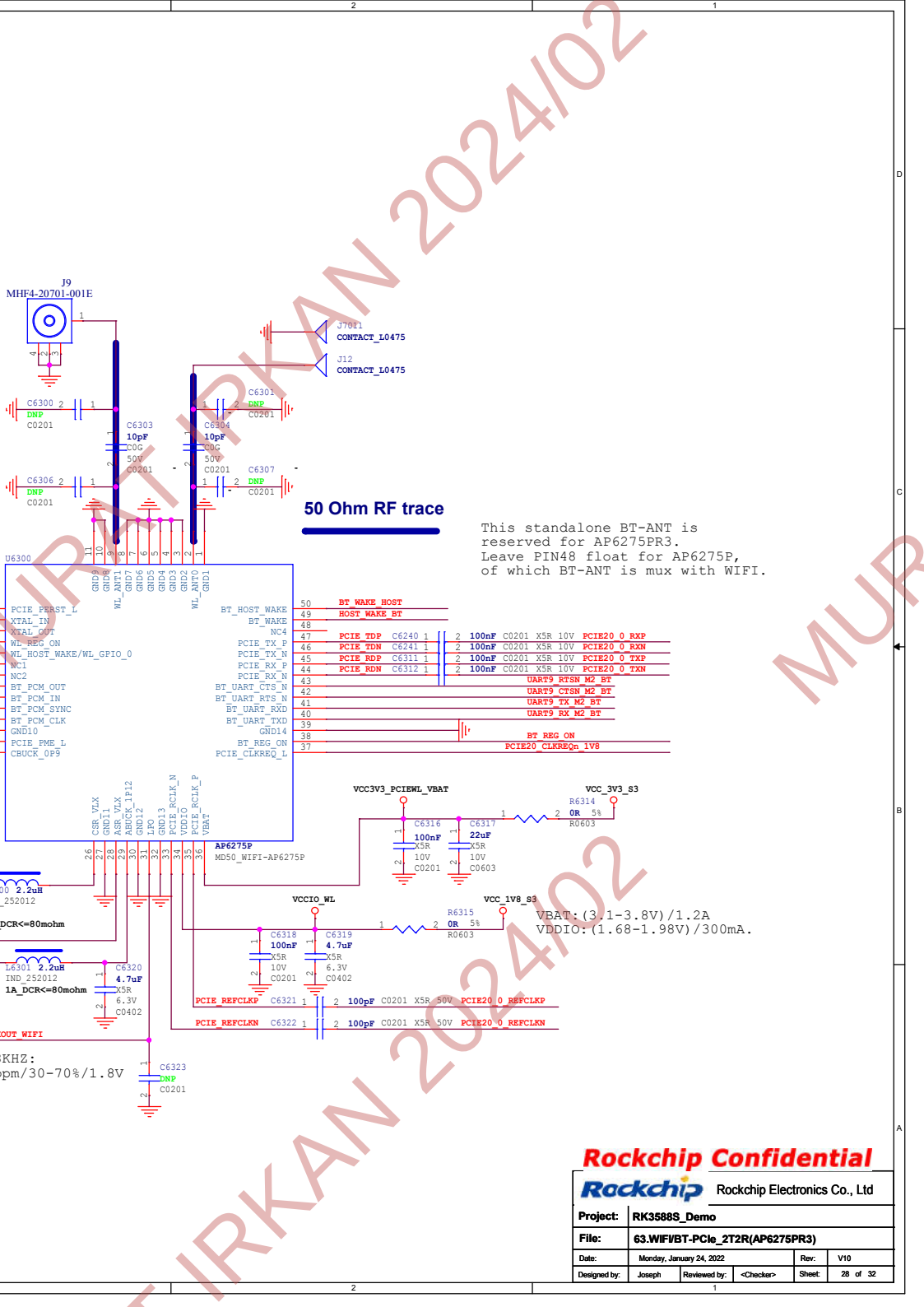


J5500
FH35C-61S-0_3SHW
CON SMD FH35C-61S-0_3SHW

The schematic diagram illustrates the RF section of the RK3588S Demo board, specifically the BT-ANT connection. The diagram shows the connection of the BT-ANT to the AP6275P module. Key components and connections include:

- BT-ANT:** A standalone BT-ANT is reserved for AP6275PR3. It is connected to the AP6275P module via a 50 Ohm RF trace. The antenna is connected to the BT-ANT pin of the AP6275P module.
- Component Values:** Various capacitors (C6300, C6301, C6302, C6303, C6304, C6305, C6306, C6307, C6308, C6309, C6310, C6311, C6312, C6313, C6314, C6315, C6316, C6317, C6318, C6319, C6320, C6321, C6322, C6323) and resistors (R6301, R6302, R6303, R6304, R6305, R6306, R6307, R6308, R6309, R6310, R6311, R6312, R6313, R6314, R6315, R6316, R6317, R6318, R6319, R6320, R6321, R6322, R6323) are shown with their respective values.
- Pin Connections:** The diagram shows the connection of the BT-ANT to the AP6275P module via a 50 Ohm RF trace. The antenna is connected to the BT-ANT pin of the AP6275P module.
- Notes:** The diagram includes a note stating: "This standalone BT-ANT is reserved for AP6275PR3. Leave PIN48 float for AP6275P, of which BT-ANT is mux with WIFI."

The diagram is a detailed schematic of the RF section of the RK3588S Demo board, showing the BT-ANT connection, component values, and pin connections. It is a technical drawing intended for use by engineers and technicians working on the board.



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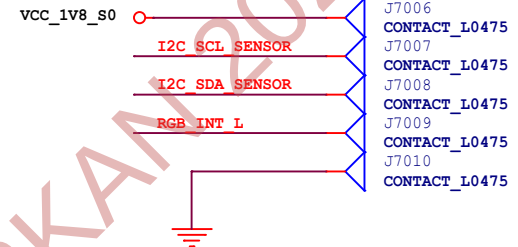
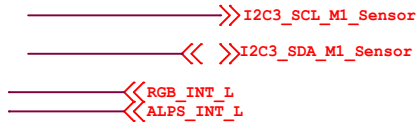
The schematic diagram illustrates the RF section of the RK3588S Demo board, specifically the BT-ANT connection. The diagram shows the connection of the BT-ANT to the AP6275P module. Key components and connections include:

- BT-ANT:** A standalone BT-ANT is reserved for AP6275PR3. It is connected to the AP6275P module via a 50 Ohm RF trace. The antenna is connected to the BT-ANT pin of the AP6275P module.
- Component Values:** Various capacitors (C6300, C6301, C6302, C6303, C6304, C6305, C6306, C6307, C6308, C6309, C6310, C6311, C6312, C6313, C6314, C6315, C6316, C6317, C6318, C6319, C6320, C6321, C6322, C6323) and resistors (R6301, R6302, R6303, R6304, R6305, R6306, R6307, R6308, R6309, R6310, R6311, R6312, R6313, R6314, R6315, R6316, R6317, R6318, R6319, R6320, R6321, R6322, R6323) are shown with their respective values.
- Pin Connections:** The diagram shows the connection of the BT-ANT to the AP6275P module via a 50 Ohm RF trace. The antenna is connected to the BT-ANT pin of the AP6275P module.
- Notes:** The diagram includes a note stating: "This standalone BT-ANT is reserved for AP6275PR3. Leave PIN48 float for AP6275P, of which BT-ANT is mux with WIFI."

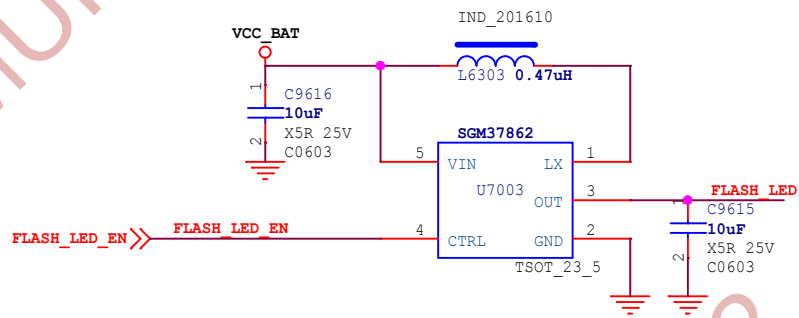
The diagram is a detailed schematic of the RF section of the RK3588S Demo board, showing the BT-ANT connection, component values, and pin connections. It is a technical drawing intended for use by engineers and technicians working on the board.

[illegible]

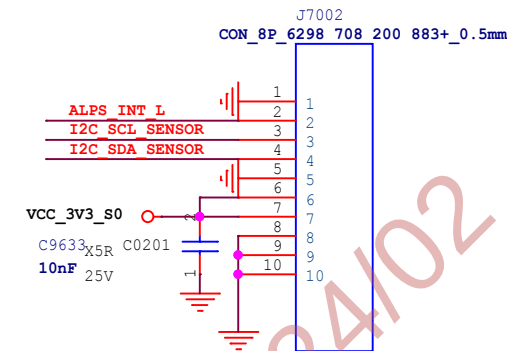
Sensor



PLS+ALS



MAX: 1.5A

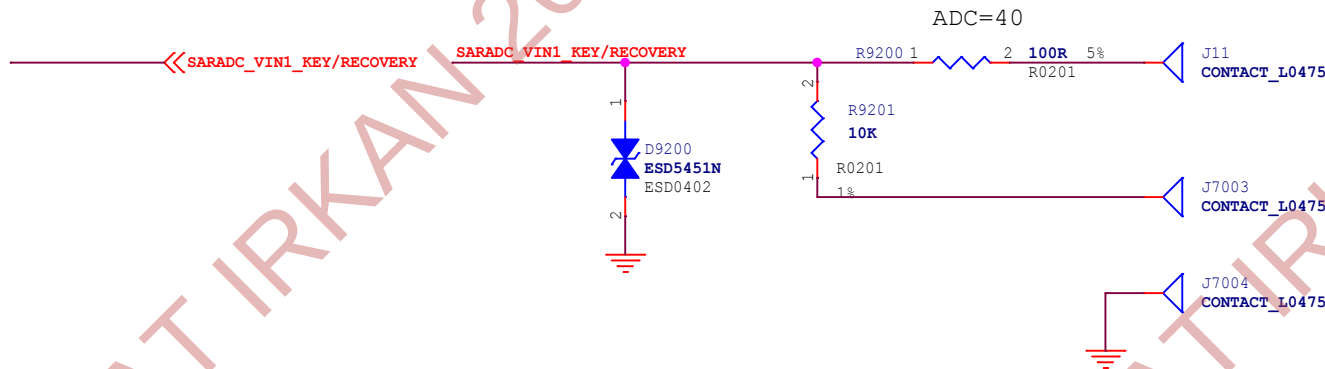


Flashlight

Rockchip Confidential

		Rockchip Electronics Co., Ltd	
Project:	RK3588S_Demo		
File:	90.Sensor		
Date:	Friday, January 07, 2022		Rev: V10
Designed by:	Joseph	Reviewed by:	<Checker>
		Sheet:	30 of 32

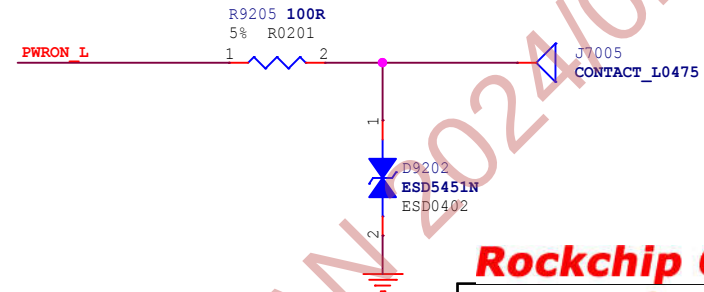
KEY Array



Reset_Key

RESET_L
PWRON_L

PWR_Key



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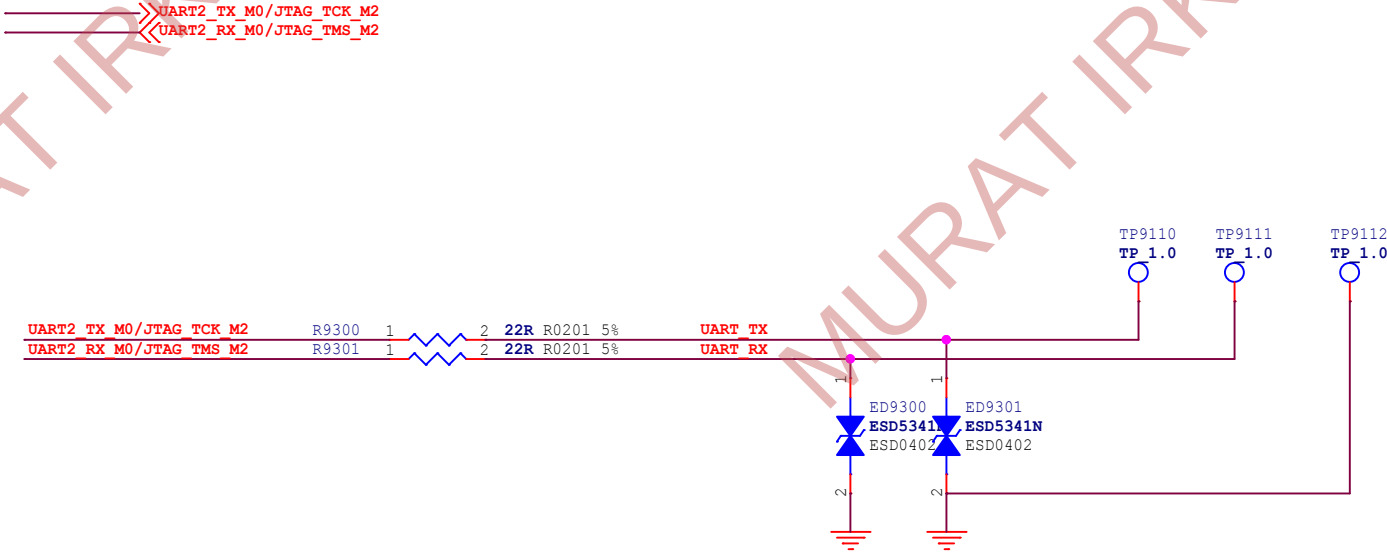


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Project:	RK3588S_Demo				
File:	92.KEY Array				
Date:	Friday, January 07, 2022			Rev:	V10
Designed by:	Joseph	Reviewed by:	<Checker>	Sheet:	31 of 32

UART Debug

JTAG Debug



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Project:	RK3588S_Demo				
File:	93.Debug UART/JTAG Port				
Date:	Friday, January 07, 2022			Rev:	V10
Designed by:	Joseph	Reviewed by:	<Checker>	Sheet:	32 of 32